

# Portable Spec Kit — Spec-Persistent Development

Technical Overview — Architecture Reference Document

A comprehensive technical document covering the framework's architecture, user profile system, flow documentation, agent compatibility, security model, testing infrastructure, and deployment strategy.

v0.6.64 • May 2026

**Author:** Dr. Aqib Mumtaz, Ph.D.

**Specialization:** Computer Science — Artificial Intelligence

**Version:** v0.6.64

**Tests:** 2865 passing (2720 framework + 145 benchmarking), 102 sections

# Executive Summary

**Portable Spec Kit** is a lightweight, zero-install, personalized spec-persistent development framework for AI-assisted engineering. It consists of a single markdown file ( `portable-spec-kit.md` ) that any AI coding agent reads to follow consistent engineering standards, manage specifications, track tasks, and maintain context across sessions.

**Key numbers:** 1 file, 5 AI agents supported, 10 core strengths, 8 source code templates, 30 flow documents, 2865 automated tests (2720 framework + 145 benchmarking), 102 test sections.

## Key Highlights

| Metric              | Value                                                                                                                                                                                                                                                                          |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Current version     | v0.6.64                                                                                                                                                                                                                                                                        |
| Framework size      | Single markdown file (approx. 980 lines)                                                                                                                                                                                                                                       |
| Skill files         | 25 (loaded JIT at peak attention)                                                                                                                                                                                                                                              |
| Reliability scripts | 29 (sync-check, critic-spawn, release, install-hooks, uninstall, code-review, scope-check, Jira, tracker, orchestrate, version-cascade, generate-ci, generate-user-guide, and more)                                                                                            |
| Enforcement layers  | 3 (bash critic, sub-agent critic, git/Claude Code hooks)                                                                                                                                                                                                                       |
| Dependencies        | Zero                                                                                                                                                                                                                                                                           |
| Install time        | Seconds (one curl command)                                                                                                                                                                                                                                                     |
| AI agents supported | Claude, Copilot, Cursor, Windsurf, Cline                                                                                                                                                                                                                                       |
| Core strengths      | 16 (Lightweight, Portable, Personalized, Spec-Persistent, Agent-Agnostic, Context-Persistent, Secure, Non-Blocking, Self-Scaffolding, Self-Validating, Automated Test Generation, CI/CD Ready, Reliably Enforced, Sub-Agent Honesty Gate, Security Scanning, Bypass Audit Log) |

|                       |                                                                                                                                                                                   |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Source code templates | 10 (Next.js Full-Stack colocated, Next.js Frontend-Only SPA, Next.js Monorepo, Python Backend, Full Stack separate services, Full Stack + Mobile, Mobile, Android, iOS, Document) |
| Automated tests       | 2351 (2206 framework + 145 benchmarking)                                                                                                                                          |
| Research              | SPD concept paper (5,300 words), benchmarking report, agentic communication discovery                                                                                             |
| Flow documents        | 22 step-by-step work flows                                                                                                                                                        |
| Project scenarios     | 9 (new, existing, monorepo, partial, cloned, subdir, returning, multiple, kit in one dir)                                                                                         |
| Profile setup         | One-time, GitHub auto-detect, 3 preference questions, per-user workspace files                                                                                                    |
| Versioning            | Two-level: framework v0.x.x (patches) + release v0.x (milestones)                                                                                                                 |

# Version Changelog

---

## v0.6 — AVACR Adversarial Framing + Sandbox Worktree + Peer-Exchange (April 2026)

Kit: v0.6.0 — v0.6.64

- **v0.6.64 — Kit Fidelity (8th Reliability Layer) (May 2026):** Closes the recurring "agent silently deviates from canonical kit commands" failure class — the same trust-based failure mode §Spawn Fidelity closed for sub-agent spawns in v0.6.60, repeated one level up at the operator-command surface. **Added:** §Kit Fidelity in `portable-spec-kit.md` with two principles: (1) *canonical default form* — every kit command runs in its canonical default form; non-canonical variants require `--rationale "<text>"` (≥20 chars) committed to `agent/.kit-deviation-log`; (2) *friction = kit bug* — workarounds are forbidden; the agent's first action on friction is to file a `KIT-GAP-NNNN` in `agent/.kit-gap-log`, then either fix the kit inline or escalate to user. **New wrapper** `agent/scripts/psk-kit-cmd.sh` with no inline-fallback branch. Subcommands: `reflex` | `prepare-release` | `refresh-release` | `init` | `orchestrate` | `feature-complete` | `new-setup` | `existing-setup` | `run-plan`. **New canonical-command inventory** `.portable-spec-kit/kit-commands.yml` (schema\_version: 1, data-driven extension). **New sync-check rule PSK040** in `agent/scripts/psk-sync-check.sh` — defense-in-depth audit of git log for marker-shaped commits without matching deviation-log entries. **New flow doc** `docs/work-flows/30-kit-fidelity.md` + **JIT-loaded skill** `.portable-spec-kit/skills/kit-fidelity.md` + **test section** `tests/sections/97-kit-fidelity.sh` (33 assertions wired into orchestrator) + **ADR-089** in `agent/PLANS.md`. **Changed:** §Reliability Architecture overview text bumped from "six enforcement layers" to "eight enforcement layers" (the 7th — §Workflow Declaration — was shipped in v0.6.62 but the overview text wasn't synced). **Tests:** 2865 (2720 framework + 145 benchmarking), 0 failures; 33 new regression tests in §97; 30 flow documents (added #30). Plan: `agent/plans/2026-05-31-kit-fidelity-8th-layer.md` (status: done, commits 45bb4cb7..9cd4f565).
- **v0.6.63 — Reflex Convergence + Workflow Declaration Flow Doc (May 2026):** Closes the v0.6.62 follow-up — every QA-Agent finding from reflex cycle-22 fixed at the kit-machinery level, plus the documentation gap for v0.6.62's 7th reliability layer. **Fixed:** QA-C22-04 CRITICAL — `findings-registry.sh` honors `REFLEX_FINDINGS_REGISTRY` + `REFLEX_HISTORY_DIR` env overrides so Section 84 tests no longer mutate the committed registry (deterministic suite); QA-C22-07 CRITICAL — `update-eval-trace.sh` emits the "Cycle-numbering note" callout in the generated header so per-pass regeneration is deterministic (test 82.19 stable); QA-C22-09 MINOR — PSK032 grandfather exemption for cycles carrying `migration-note.md` (documented historical mis-numbering) plus regression test 82.21; QA-C22-08 MINOR — `doc-code-diff.sh` honors `# doc-coverage-exempt:` header marker applied to 4 internal mechanical helpers ( `psk-`

behavior-live , psk-behavior-parity , psk-gen-manifest , psk-state-cleanup ), precise (only marked scripts opt out — no blanket silencing), 2 new regression tests in 04-reflex.sh . **Added:** docs/work-flows/29-workflow-declaration-schema.md documents the 7th reliability layer introduced in v0.6.62 (PSK034/PSK035 + Class A/B/B'/B-plan-driver/C taxonomy + phases.yml schema + dispatch contract). **Changed:** stale reinit references in 13-release-workflow.md repointed — 5 executable workflows (release · feature-complete · init · new-setup · existing-setup) with reinit documented as folded into idempotent init . **Tests:** 2832 (2687 framework + 145 benchmarking), 0 failures; 3 new regression tests (82.21 PSK032 grandfather + 2 doc-coverage-exempt in 04-reflex.sh); 30 flow documents (29 user-facing + 1 template).

- **v0.6.62 — Command-Model Redesign + Dispatcher Migration (May 2026):** Collapses the kit to a clean idempotent command model and completes the dispatcher migration. **Added:** Two-command model — idempotent install (pull machinery) + init (registry-driven conformance engine, CREATE-or-REFRESH, content-loss-protected, folds the retired reinit via psk-reinit.sh thin alias). Unified orchestrate build — one workflow for new + existing projects (each phase idempotent create-or-update). PSK034 + PSK035 sync-check rules implemented (were documented but never coded) — workflow-router↔phases.yml mapping + phases.yml schema validation. # workflow-decl-exempt: markers on plan-driver/session/gate helpers (Class B-plan-driver). **Changed:** psk-run-plan.sh delegates start/next/resume/retry to psk-dispatch.sh --plan — every plan spawn now routes through psk-spawn.sh (closes the bare- SPAWN: -echo §Spawn-Fidelity gap); compat-mode execution ported into the dispatcher; absolute-artifact-path robustness added. Class A/B/B'/B-plan-driver/C taxonomy formalised. **Removed:** --update / --retrofit orchestrator modes (existing-project structural conformance is init 's separate axis); separate reinit / reinstall / new-project verbs (consolidated into idempotent install + init + build ). **Resolved:** reflex as monolithic-by-design (Class B') — its iterate-until-convergence control flow does not fit the dispatcher's linear phases.yml model; already §Spawn-Fidelity-compliant via spawn-qa.sh / spawn-dev.sh → psk-spawn.sh . **Tests:** 2832 (2687 framework + 145 benchmarking), 0 failures; Section 89 added (11 regression tests for PSK034/PSK035 workflow-declaration enforcement); 29 flow documents.
- **v0.6.61 — Strict-GRANT Convergence (May 2026):** Closes the convergence gap between v0.6.60 mechanical shipping and v0.6.60 strict-GRANT verdict. **Added:** PSK031 cross-pass findings registry ( reflex/lib/findings-registry.sh ~450 LOC + reflex/history/findings-registry.yaml ) — content-fingerprint canonical-ID tracking with widened/residual alias support; bootstraps 89 canonical IDs from cycle-17/18/20. PSK032 cycle-numbering misuse detection — fires when consecutive cycle dirs have only pass-001 with verdict.md present (cycle-tracking bypassed). PSK033 standalone-pass overuse advisory. Watchdog hooks wired into reflex/run.sh Step 1b + reflex/lib/preconditions.sh Gate 5 + session-start ( psk-resume-bootstrap.sh Step 4). psk-workflow-state.sh prune subcommand for workflow-state retention. New flow-doc sections in 17-reflex.md (§Cycle vs Pass semantics, §Standalone pass-dir layout, §Watchdog Hooks, §Findings Registry). **Changed:** reflex/lib/check-audit-

`completeness.sh` `wall_clock` multi-wave aggregation (6 fallback strategies; cycle-17/18/20 verdicts `suspect` → `real` post-fix); `file-bugs.sh` integrated with findings-registry dedup; `dev-self-verify.sh` `eval` → `bash -c` hardening; `psk-release.sh` declares `script-class: orchestrator` (43/43 `psk-*` scripts now classified). **Fixed:** QA-D4-03 MAJOR (workflow watchdog HF4b now structurally invoked across passes); QA-D6-03 MAJOR (`wall_clock` multi-wave aggregator bug); cycle-numbering bug in `compute_next_cycle_id()` / `next_cycle_id()` (KV-vs-YAML parser mismatch — each reflex invocation created new cycle-NN instead of incrementing pass-NNN). **Convergence:** cycle-21 STRICT GRANT (0 MAJOR/CRITICAL/MINOR open, 2 POSITIVE ADVISORY only; trajectory 53→30→0 open findings across cycle-17→cycle-20→cycle-21). **Tests:** 2658 (2513 framework + 145 benchmarking); Sections 85/86/87 added (P4/P5/P6 regression suites, 98 new tests); 33 sync-check rules (PSK001-PSK033); 84 features, 100% R→F→T coverage; 29 flow documents.

- **v0.6.60 — Spawn Fidelity Hardening (May 2026):** Structural prevention of synthesis-as-shortcut. After the v0.6.59 reflex incident where SDK stream-idle-timeouts caused inline-synthesis fallback, this release establishes **§Spawn Fidelity** as the 6th reliability layer with 9 enforcement mechanisms. **Added:** §Spawn Fidelity rule in `portable-spec-kit.md` + P11 principle in `agent/PHILOSOPHY.md` ; Standard Spawn Recipe skill ( `.portable-spec-kit/skills/spawn-fidelity.md` ); persistent retry queue ( `agent/.workflow-state/retry-queue.yml` + `psk-retry-queue.sh` ); resume-on-session-start MANDATORY rule ( `psk-resume-bootstrap.sh` + PSK029); workflow watchdog ( `psk-workflow-watchdog.sh` ) + phase idempotency rule; synthesis-detection probe ( `reflex/lib/check-audit-completeness.sh` ); 13th mechanical gate `audit-completeness` in `reflex/lib/gates.sh` + PSK026 critic-completeness sync-check; Dim 27 Synthesis-Detection + Dim 28 Spawn-Coverage Audit in `reflex/prompts/qa-agent.md` ; PSK027 bypass-tamper-detection ( `psk-bypass-log.sh` ); flow doc 28 ( `28-spawn-fidelity.md` ). **Changed:** reflex `spawn-qa.sh` + `spawn-dev.sh` retrofitted through `psk-spawn.sh` (HF1); `psk-critic-spawn.sh` retrofitted (HF2, all 6 critic templates); `psk-orchestrate.sh --update` U3-U10 split from single batch into per-U-phase spawns (HF1b — closes 15h-batch context-overflow risk); §Reliability Architecture opening updated from "five enforcement layers" → "six enforcement layers". **Removed:** cascade-as-user-update path purged from 5 kit sites — canonical user-project update is now `bash agent/scripts/psk-orchestrate.sh --update` ; PSK028 anti-pattern sync-check flags reintroduction. **Tests:** 2522 (2377 framework + 145 benchmarking); reliability suite +156 PASS tests across Sections 68-81 (433 → 589); 30/30 sync-check rules; 84 features (F77-F84 added), 100% R→F→T coverage; 28 flow documents.
- **v0.6.59 — Workflow-Fidelity Plan Close-out (Phase D, May 2026):** Hygiene release sweeping Phase R's 4 ADVISORY findings + PSK022a polyglot-detection word-boundary regex fix. Plan schema conversion completed for both `workflow-fidelity.md` (22 phases) and `plan-save-protocol.md` (7 phases) from `compat_mode` → `schema_version: 1` via `psk-run-plan.sh --convert .` `ard/user-guide.html` regenerated and present. `install.sh` inventory swept to include 6 new artifacts.

**PSK022a root cause:** the polyglot-detection regex matched `Gin` inside unrelated tokens like `messAGING` and `enGINE` without word boundaries — fixed by anchoring every keyword as `\b(Keyword)\b` and excluding library names (Drizzle, Twilio, Upstash, NextAuth) from runtime detection. v5 Next.js + Drizzle + Twilio + Neon stack now correctly resolves as Template 1 (single-runtime colocated) instead of being mis-classified as polyglot. New SPECS features F73-F76 added covering schema conversions, user-guide regen, install.sh inventory, and PSK022a fix. 2325 tests passing (2180 framework + 145 benchmarking), 76 features, 100% R→F→T coverage.

- **v0.6.57 — Plan Execution Protocol + UI Completeness Gate (May 2026):** New MANDATORY §Plan Execution Protocol in `portable-spec-kit.md` — the 5th reliability layer (plan-shape gate). Every plan executed via sub-agent spawning conforms to `schema_version: 1` + `phases:` frontmatter. New `agent/scripts/psk-run-plan.sh` (35th script) — schema-aware driver with commands `start/next/status/resume/retry/--convert/--validate/--health/abort`; structurally no inline-fallback branch (only `spawn/retry/abort` paths forward). Three new templates (`plan-executable.md` + `plan-prompt.md` + `plan-artifact.md`) all pass the 7-criterion Template Quality Bar. `psk-plan-save.sh` body-preservation fix (today's regression erased 233 lines of plan content; fixed via `tempfile` + `frontmatter-only` rewrite) + schema validation on `approve` (PSK024 codes N/P/F/D/L). PSK024 sync-check rule (27th rule, 11 sub-codes V/P/I/N/R/A/G/C/D/M/X) lints every `agent/plans/*.md`. §Workflow Fidelity B2 retrofit: `register-gate` / `verify-gate` API on `psk-workflow-state.sh` + 9 workflows wired (`mark-done` refuses unverified phases). New `agent/scripts/psk-ui-completeness.sh` (36th script) — 10-category stack-aware UI audit (P primitives, L layout, D dark-mode, S states, A a11y, T tokens, F per-feature pages, I input-feedback, R responsive, E empty-shell), bash 3.2 compatible. PSK025 sync-check rule (28th rule). Orchestrator `--update` U6.5 — UI-completeness backfill phase that scaffolds missing primitives/layout/dark-mode/states/a11y/tokens/pages until `--strict` exits 0; no skip path. Reflex Dim 26 + 12th mechanical gate `workflow-fidelity-completeness`. New flow docs 26 (Plan Execution Protocol) + 27 (UI Completeness Gate). New `.portable-spec-kit/skills/plan-execution.md`.  
**Motivating incidents:** (1) `psk-plan-save.sh save` regression that erased plan body content — fixed in B0.3 with body-preservation contract. (2) `searchsocialtruth-v5` skeletal UI — empty-shell pages now auto-flagged as PSK025-E. 2325 tests (2180+145), 390 reliability tests across Sections 61-67.
- **v0.6.56 — Workflow Fidelity: the agent executes the kit, never overrides it (May 2026):** New MANDATORY §Workflow Fidelity section in `portable-spec-kit.md` — the 4th reliability layer, behavioral and structurally enforced. When the kit defines a process (workflow, phase, gate, rule, principle, test) the agent executes it faithfully and completely. Rate/context limits → `pause-and-resume`, never `reduce-scope`. New principle `P10` — Workflow Fidelity in `agent/PHILOSOPHY.md`. New `agent/scripts/psk-workflow-state.sh` (33rd script) + `agent/scripts/psk-spawn.sh` (34th script). 9 executable workflows retrofitted. New flow doc `25-workflow-fidelity.md`. 18 regression tests (section 61). Motivating incident: `searchsocialtruth-v5` shipped a skeletal UI behind a "20/20 features, 110 tests" report.

- v0.6.55 — Source Layout Policy + 10 new templates + flow doc 24 (May 2026):** Stack-driven source-layout policy — orchestrator reads `agent/PLANS.md` Stack table → derives Template choice (1 colocated / 3 separate-services / 4 monorepo / 5 library / 6 CLI) + opt-in `src/` subdirs (`core` + `shared` always; `ui` / `api` / `integrations` / `platform` per Stack). New `psk-scaffold-src.sh`. PSK022a (template choice) + PSK022b (src subdir layout) sync-check rules. 10 new file-class templates (23 total). Flow doc 24 (template-driven creation). 2352 tests (2207+145).
- v0.6.54 — Template Quality Bar (May 2026):** New §Template Quality Bar rule — 7 mandatory criteria (stack-agnostic, domain-agnostic, scale-agnostic, useful-and-complete, lifecycle-aware, self-documenting, round-trippable). `.portable-spec-kit/templates/` externalized templates dir (13 templates extracted from inline). `agent/scripts/psk-template-quality.sh` — 7-criterion lint with JSON output + `--strict` gate mode. PSK023 sync-check + `.audit-log.yaml`. 2337 tests (2192+145).
- v0.6.53 — Plan-Save Protocol: interruption-resilient plan persistence (May 2026):** New mandatory rule in `portable-spec-kit.md` §Task Tracking — conversation-drafted plans persist to `agent/plans/YYYY-MM-DD-<slug>.md` as the agent's first tool call, before any implementation. Lifecycle frontmatter drives `draft` → `approved` → `executing` → `done` | `abandoned`. New `agent/scripts/psk-plan-save.sh` (30th script). Flow doc 16 updated. Closes the IDE-ephemeral plan-storage gap. 2337 tests (2192+145).
- v0.6.52 — Reflex cycle-10/pass-003 close: ARD content correction + doc-code-diff \*/basename fix + flow doc count sweep (May 2026):** ARD Technical Overview corrected: v0.6.51 section had wrong content (PSK019/workflow-template-standard from v0.6.50 was showing under v0.6.51 heading); replaced with correct pass-002 merge content; missing v0.6.50 entry added. `doc-code-diff.sh` `*/basename` pattern fix eliminates 4 false-positive MAJORS per pass; +1 framework test (2191→2192). Test count sweep: 8 files updated 2336/2191 → 2337/2192. Flow doc corrections in docs 17, 19, 21. 2351 tests (2206+145).
- v0.6.51 — Reflex cycle-10/pass-002 merge: doc-code-diff fixes + F16 test hardening + K3/K7 cross-refs (May 2026):** Fast-forward merge of `reflex/dev-cycle-10-pass-002` into main — 119 files committed (`findings.yaml`, `signoff.md`, `verdict.md`, `dev-trace.md`, 72 F-test plans in `behavioral-tests/`, `coverage.md`, `gates-result.md`, `.cycle-meta`). **doc-code-diff.sh template exclusions:** Added false-positive filter patterns for placeholder file names used in kit documentation examples: `foo.sh`, `psk-foo.sh`, `script-name.sh`, `test_f*_feature_name.*`, `f*-feature-name.*` — eliminates spurious MAJOR noise from instructional literals. **F16 test hardened (exact count):** `tests/features/f16-work-flow-docs.sh` assertion changed from `>=` to `==`; `00-template.md` explicitly excluded — test now asserts exactly 23 numbered docs. **23-project-update-workflow.md K3/K7 additions:** Added K3 (design-plan + stub creation phases U3/U4), enhanced K6 cross-refs, added K7 (`HANDOFF.md` structure for U10). **AGENT\_CONTEXT.md test count corrected:** Stale count 1982 replaced with 2336 (2191+145). 2351 tests (2206+145 — +1 from doc-code-diff basename fix).
- v0.6.50 — Workflow template standard: PSK019 gate + all 23 flow docs standardised + Gate 1 scope fix (May 2026):** New

`docs/work-flows/00-template.md` establishes the canonical 7-section workflow template with 63-char ASCII box standard (Overview, Flow Diagram, Steps, Key Rules, Edge Cases, Related Flows, Bypass). **PSK019 gate** added to `psk-sync-check.sh` — `check_flow_doc_template()` enforces `## Overview`, `## Flow Diagram`, and `## Key Rules` in every flow doc except the template itself. All 23 existing flow docs updated to comply; PSK015 and PSK003 exclude `00-template.md` from doc-count and file-count checks. Box line alignment: all ASCII box lines fixed to exactly 63 display characters — 80 misaligned lines corrected across 13 files. **Reflex Gate 1 scope fix:** `reflex/lib/preconditions.sh` Gate 1 changed from `git status --porcelain` to `git status --porcelain -- .` — scopes the clean-tree check to `PROJ_ROOT` subtree only, preventing false positives when the kit lives inside a larger workspace git repo with sibling untracked directories. 2336 tests (2191+145).

- **v0.6.49 — resolve\_kit\_root bug fix: path-spaces stripping + heuristic false-positive (May 2026):** Two bugs in `psk-orchestrate.sh resolve_kit_root()` fixed. (1) `tr -d '[:space:]'` stripped embedded spaces from directory names — fixed with `sed 's/[[:space:]]*$/'`. (2) Heuristic false-positive: any project with a real copy of `portable-spec-kit.md` identified as kit root — fixed by adding `[ -f "$kit_candidate/install.sh" ]` check. 2336 tests (2191+145).
- **v0.6.48 — Orchestrator --update mode + Phase 9 mandatory reflex + KIT\_ROOT resolution + polyglot stack detection + R→F→T stubs + summary.csv GRANTED path + project-update flow doc (May 2026):** K1: `install.sh` writes `kit_source_path / kit_version / kit_installed_at` to `.portable-spec-kit/config.md`. K2: `psk-orchestrate.sh` Phase 9 reflex-audit now MANDATORY — no skip path, auto-installs reflex if missing. K3: `psk-orchestrate.sh --update` mode (U0–U10 phases: script sync → mandate audit → reflex install → design plans → feature stubs → ARD/flows → sync-check config → features → release-prep → reflex GRANTED → handoff). K4: Phase 7 sub-agent prompt updated to require `tests/features/` R→F→T anchor stubs. K5: `mandate-audit.sh` polyglot detection added — `backend/+frontend/` projects resolve as `polyglot-python-node` instead of `unknown`. K6: `loop.sh` GRANTED terminator calls `score.sh`; cycle-09 backfilled in `summary.csv`. K7: Flow doc 23 ( `23-project-update-workflow.md` ) retires manual cascade pattern. 2336 tests (2191+145).
- **v0.6.44 — Kit-evolution phases 1–6: config.md, template enrichment, F71/F72, test infra fixes (May 2026):** `.portable-spec-kit/config.md` created + wired into `install.sh` + PSK015-CONFIG gate + MANDATE-CONFIG-MISSING probe + T4/T5 regression tests. All 9 agent file templates enriched (approval tracking, scope-change record, Security, Definition of Done, QA Findings, Evidence, Open Research Questions). F71 (6 tests) + F72 (15 tests) regression suites added. PSK016 stub-path gate + Dim 25 probes 25.2–25.4 added. Per-feature test loop fixed: SCRIPT\_DIR clobber + cwd deletion bugs; 72 feature files now run. 2336 tests (2191+145).
- **v0.6.43 — Flow doc consistency sweep: step counts, gate count, dimension count, test count (May 2026):** `11-spec-persistent-development.md` : "8-step gate" → "10-step gate" (two occurrences). `17-reflex.md` : "10 gates" → "11 gates" (PKFL added G1 kit-evolution-file-scope + G2 kit-genericity-proof in v0.6.41). `18-project-`

orchestration.md : "24 dimensions" → "25 dimensions" (Dim 25 Mandate-Compliance added in Loop-Iter-2). 19-kit-evolution-gauntlet.md : "1836 tests" → "1982 tests". Test infrastructure: flow count check 21→22. 1982 tests (1837+145).

- **v0.6.42 — README consistency sweep: script count + flow table row 22 (May 2026):** PSK013 fix: Quick Start "Installs:" updated from 27 → 29 scripts (psk-generate-ci.sh + psk-generate-user-guide.sh added in v0.6.41). PSK015 fix: Flows table row 22 added for 22-project-kit-feedback-loop.md ; Documentation section updated to 22 flow diagrams. ARD Key Highlights updated: flow docs 21→22, scripts 27→29. 1982 tests (1837+145).
- **v0.6.41 — PKFL + convergence discipline + ARD user guide + mandate audit extensions (May 2026):** Convergence discipline fix (critical): Rule 4 in loop.sh + run.sh — DENIED + 0 unclosed findings now stays in same cycle (was falsely advancing). **PKFL machinery:** reflex/lib/kit-evolution.sh (6-phase orchestrator), check-kit-genericity.sh (G2 vocabulary scanner), smoke-test-examples.sh (G6 multi-project smoke); gates 10 (G1 file-scope) + 11 (G2 genericity-proof) added to gates.sh ; G4 genericity\_proof gate in file-bugs.sh ; flow doc 22-project-kit-feedback-loop.md created; §PKFL mandatory rule section added to portable-spec-kit.md ; gates count 9 → 11. **ARD user guide generation:** psk-generate-user-guide.sh + psk-generate-ci.sh , wired into psk-release.sh Steps 1 + 7. **Dim 25 mandate audit extensions:** ARD user guide absent (ADVISORY), DB migration scaffold absent (MINOR), shadcn/ui components empty (MINOR). Stack-specific fix patterns added to dev-agent.md . 1982 tests (1837+145).
- **v0.6.40 — Flow doc coverage for v0.6.39 features (May 2026):** Prep-release flow doc updates: 17-reflex.md gains §v0.6.39 capabilities section (server-lifecycle.sh, Next.js colocated template, template-strict scaffold, cycle persistence, cascade Kit field); 18-project-orchestration.md Phase 6 scaffold row updated to reference template-strict Stack matching rule. 1982 tests (1837+145).
- **v0.6.39 — Playwright server-lifecycle, Next.js colocated template, template-strict scaffold, cycle persistence, cascade Kit field (May 2026):** Five structural improvements. **server-lifecycle.sh (NEW):** starts dev server before QA-Agent; port poll via nc -z ; PID to .server.pid ; process group kill for webpack/esbuild. spawn-qa.sh start hook + run.sh stop hook + reflex/config.yml server\_lifecycle: block ( enabled: false default) + qa-agent.md REFLEX\_SERVER\_STATUS awareness (ADR-091). **source-structures.md:** Next.js Full-Stack (colocated) DEFAULT template — ALL code under src/ , public/ at root (framework constraint), drizzle.config.ts out: ./src/db/migrations ; Stack matching rule; Monorepo template added. **psk-orchestrate.sh Phase 6:** template-strict scaffold reads PLANS.md Stack → matches template → creates dirs exactly per template. **Single-mode cycle persistence (reflex/run.sh):** .active-cycle state file persists cycle-id across bash invocations; cleared on GRANTED. **Cascade Kit field (psk-version-cascade.sh Phase Q):** updates \*\*Kit:\*\* field in sync-target AGENT\_CONTEXT.md. 1982 tests (1837+145).
- **v0.6.38 — Kit-quality fixes: playwright gate, verdict integrity, findings.yaml discipline, flow doc coverage (May 2026):** 9 targeted kit fixes from searchsocialtruth

cycle-07 kit-scope findings (kit-first philosophy). Playwright gate now runs `npm ci --prefer-offline --silent` before `npx playwright test` + adds `playwright_suite_gate_enabled` config key. `write_verdict()` in `run.sh` emits explicit `verdict: GRANTED|DENIED` field for the convergence-audit gate. `dev-agent.md` mandates `findings.yaml` status `open→closed` after `TASKS.md` [x] + promotes `[source: <pass-name>]` to HARD REQUIREMENT. `track-tokens.sh` unbound variable fix. `psk-sync-check.sh` new `check_kit_version_drift()` advisory. `.gitignore` gains playwright output paths. `mandate-audit.sh` emits ADVISORY when both root `REQUIREMENTS.md` and `agent/REQS.md` exist. `psk-orchestrate.sh` Phase 7 explicitly requires `tests/f{N}-feature-name.test.ts` naming. `REQUIREMENTS.md` deleted from kit root — `agent/REQS.md` is authoritative. Flow doc coverage: 13-release-workflow.md Step 6 + 17-reflex.md §v0.6.30/§v0.6.35 sections. 1982 tests (1837+145).

- v0.6.0 — F70 Reflex evolves to AVACR (Adversarial Verbal Actor-Critic Refinement Loop):** Architectural shift within F70 Reflex from cooperative VACR to adversarial goals with collaborative convergence. *QA-Agent* goal = FAIL the release (hunt exhaustively across 14 dimensions; default DENIED). *Dev-Agent* goal = FOOLPROOF the release against QA's hunt (fix findings + harden sibling-class weaknesses). Convergence = QA hunted hard and cannot find a blocker, not "Dev says it's done."
  - Physical sandbox isolation:** QA operates in a git worktree at `reflex/sandbox/qa-pass-NNN/` with `agent/AGENT.md` + `agent/AGENT_CONTEXT.md` physically removed. Dev's narrative files cannot bias QA's fresh-context investigation — enforced by filesystem, not prompt discipline. Dev retains full-repo read + write. Dual-rooted access (`REFLEX_QA_WORKTREE` for reads, `REFLEX_QA_PROJECT` for invocations).
  - Peer-to-peer YAML exchange:** QA writes `findings.yaml` (per-finding: priority + dimension + blast\_radius + confidence + reproducibility + escalation + citable\_quote + regression\_vector + standard + recommendation). Dev returns `dev-result.yaml` (fixed / rebuttals / routed\_to\_human / escalated / deferred). Human reads only `signoff.md` (GRANTED or DENIED with exit criteria) — never agent working output.
  - 14-dimension adversarial hunt:** functional / edge-case / cross-pipeline / workflow / research-backed-quality / superficiality / production-readiness / security (OWASP) / performance / architectural-compliance / documentation / test-quality / tech-debt / readiness-affirmation. Combined with **4-persona testing** (new-user / power-user / malicious-user / accessibility-user) per feature.
  - Research capability:** QA may invoke WebSearch/WebFetch for RFCs, OWASP Top 10, published benchmarks, CVEs, academic papers. Paired with a **citable-quote honesty gate** — every finding requires `citable_quote.from` (file:line) + verbatim text that bash grep can verify. No hallucinated standards — `standard.url` must be real.
  - 4-bucket Dev diagnosis + Bucket-D rebuttal:** A (doc wrong) / B (code wrong) / C (both inconsistent) / D (QA misread — requires counter-citation in `dev-result.yaml` rebuttals). Dev does not dismiss findings.
  - Human-arbitration escalation:** findings tagged `escalation: human-arbitration` routed to `@<human>` ; Dev does not auto-fix cross-cutting / ADL-amendment concerns.

- **Flaky-vs-deterministic distinction:** non-reproducing findings trigger root-cause investigation (race / ordering / isolation), not patch-retry.
  - **Coverage declaration:** QA writes `coverage.md` enumerating tested × dimensions AND untested surface with reasons. Signoff covers tested only.
  - **Regression-vector re-execution:** every finding preserves exact invocation + expected assertion. Re-executed verbatim each pass. Failure upgrades priority one level (catches surface-patch vs root-cause fixes).
  - **CHANGELOG truth-check mandate:** every bullet in current version's CHANGELOG must have a test case; unverifiable claims → `QA-REL-{N}` blocks signoff.
  - **Sibling-class hardening in Dev:** on any finding, Dev audits for same-class issues across the codebase and fixes siblings as separate atomic commits. Fixing the class, not just the instance, is what separates fool-proof from whack-a-mole.
  - **Consistency sweep:** `cycle` → `pass` across ~30 references in live scripts + CSV column header. `Refine` → `Reflex` (3 pre-existing stragglers including real install/update URL bugs). `Remediation` → `Refinement`.
  - **Paper title:** *"Adversarial Verbal Actor-Critic Refinement (AVACR): Multi-Agent Verbal Reinforcement Learning with Asymmetric Goals for Spec-Grounded Post-Release Program Repair"*. Target venues: ICSE / FSE / ASE. "Adversarial" refers to inference-time asymmetric goals / red-teaming, NOT adversarial training (no gradient updates).
  - **Tests:** 1077 framework + 145 benchmarking = 1222 (unchanged count; 2 Section 60m assertions updated for AVACR formal name + black-box discipline).
  - **Bug fixes surfaced during sweep:** `reflex/install-into-project.sh` was installing into non-existent `$TARGET/refine`; `reflex/update.sh` `KIT_RAW_BASE` pointed at 404 URL. Both fixed.
- **v0.6.1 — Kit self-evolution loop + MINIMAL template tier + 14 framework-gap closures:** Large dual-track release closing the empirical-kit-evolution loop end-to-end and adding a third response-template tier for latency cuts.
- **N33 adopter-side auto-submit:** new `reflex/lib/auto-submit.sh` dispatcher at end of every pass. Three modes — *manual* (default), *prompt* (y/n with 10s timeout), *auto* (fire-and-forget). Four guards — consent marker, pass filter (verdict in {NEEDS\_FIX, REVOKED} OR kit/meta scope), 24h rate limit, anonymise. Opt-in via `bash reflex/run.sh --enable-auto-submit --i-understand-privacy-implications`. Config in `reflex/config.yml` under `upstream_submission:`. Audit log at `reflex/history/auto-submit-log.csv`.
  - **N38 maintainer-side intake automation:** new `reflex/lib/intake.sh` plus `.github/workflows/kit-intake.yml`. Weekly Monday 09:00 UTC GitHub Action fetches open `avacr-eval` issues, parses kit/meta-scope findings, drafts `agent/tasks/Gxx-<slug>.md` task files from template, opens an intake-review PR with `intake-review-pending` label. Parser is deterministic Python (no LLM on intake path). Manual run via `bash reflex/intake.sh`.

- **N44/N45 general-purpose runner + kit-loop:** `mode-less bash reflex/run.sh [--target <path>]` where kit-identity detection drives routing automatically. New `--kit-loop` chains prep-release plus self-test with max 3 iterations via state machine at `agent/.release-state/kit-loop-state.yml`. Convenience wrappers `reflex/kit-loop.sh` and `reflex/self-test.sh` (retired in v0.6.2 per entry-consolidation; use `bash reflex/run.sh` and `bash reflex/run.sh --self-test`). `--help / -h` prints usage.
- **Trace continuity across passes (QA reads prior trace):** QA-Agent prompt now mandates reading the prior pass's `findings.yaml` (structured evidence with `regression_vector.invocation_verbatim` + `expected_assertion`) and `signoff.md` (verdict + deferred decisions) at Phase 1, plus the cross-pass `REFLEX_EVAL_TRACE.md` register. `spawn-qa.sh` surfaces all three paths in the task-file preamble.
- **Per-pass canonical trace (Q1):** each pass commits `findings.yaml` (structured, one entry per finding with `citable_quote` + `regression_vector` + `recommendation`) plus `signoff.md` (verdict GRANTED / DENIED, blocking findings, deferred decisions). Cross-pass `reflex/history/REFLEX_EVAL_TRACE.md` register auto-aggregates every finding from every pass via `reflex/lib/update-eval-trace.sh` into one reviewable index (id · severity · scope · status · one-line summary). The earlier 8-section `AVACR_EVAL_TRACE.md` narrative was consolidated into this three-file model — reduces duplication across artifacts.
- **v0.6.2 — Dev-branch isolation:** every reflex pass runs Dev on a dedicated `reflex/dev-cycle-NN-pass-NNN` branch (or `reflex/dev-standalone-pass-NNN` for ad-hoc runs). User's main untouched during the pass; fast-forward merge on GRANTED + auto-delete; unhappy branches retained (last 3) for diagnosis.
- **v0.6.2 — Protected-files write-ban:** 3-layer enforcement that `agent/AGENT.md` and `agent/AGENT_CONTEXT.md` are never modified by reflex findings (prompt constraint + `gates.sh` diff check + `psk-sync-check.sh` pre-commit hook on reflex dev branches).
- **v0.6.2 — Autoloop generalization:** bare `bash reflex/run.sh` invocation is now the autoloop entry point (aliases `--autoloop / --loop / --kit-loop`) chaining prep-release + reflex pass + iterate until GRANTED, identical across kit / new project / existing project. (Earlier `reflex/autoloop.sh` wrapper retired per same-release entry-consolidation.)
- **v0.6.2 — History retention:** `reflex/lib/prune-history.sh` + `history_retention` config bound disk growth (defaults: `pass_dirs_keep 10`, `dev_branches_keep 3`, `qa_sandbox_keep 3`). Register + `summary.csv` kept forever; archived passes marked `_(archived)_` in register.
- **v0.6.2 — Concurrency + empty-pass:** `agent/.release-state/reflex.lock` prevents parallel reflex corruption; 0-finding passes shortcut to GRANTED with no Dev spawn.
- **v0.6.2 — Intake parser update:** `reflex/lib/intake.sh` reads `findings.yaml` YAML block from GitHub issue bodies (matches v0.6.2+ submission format).

- **v0.6.3 — Autonomous reflex run validation:** mechanical patch minted from an autonomous end-to-end reflex run on the v0.6.2 consolidation. QA surfaced 11 findings (3 CRITICAL including dead-code branch regex in the write-ban); Dev landed 11/11 fixes in 6 atomic commits on the dev branch, fast-forward merged to main.
- **v0.6.4 — Convergence-based stopping (N56):** replaces the naïve "max 3 iterations" cap with real convergence signals: GRANTED, REGRESSION, findings\_floor, plateau (patience\_passes), fix-rate drop (min\_fix\_rate), safety\_cap as escape hatch. Configurable in `reflex/config.yml` → `convergence:` .
- **v0.6.4 — --loop --resume collision fix:** deferred flag resolution in `run.sh` ; `--loop --resume` now routes to `loop-resume` mode instead of hijacking to `resume-qa` .
- **v0.6.4 — gates.sh cache-order bug:** rft-cache cleared at gate-runner entry so sequential gates don't see stale data from the test gate.
- **v0.6.4 — Token tracking (reflex/lib/track-tokens.sh):** per-pass tokens aggregated into `reflex/history/token-usage.csv` with `tokens_per_finding` metric + per-pass and per-cycle budget warnings.
- **v0.6.4 — Iteration auto-chain on gate-fail:** spurious gate failures no longer abort the autoloop; only REGRESSION + verdict/findings-trend signals halt.
- **v0.6.5 — Kit-bootstrap integrity gate (4-layer defense):** prevents un-installed-kit projects from running prep-release or reflex. New `agent/scripts/psk-bootstrap-check.sh` (7 structural checks C1-C7: framework file + entry-point symlink, `.portable-spec-kit/config.md` , core 6 kit scripts, cached skills, pre-commit hook, all 9 `agent/*.md` pipeline files, test harness) wired into `psk-release.sh` Step 0, `reflex/lib/preconditions.sh` Gate 0a, and QA-Agent Dimension 16. Catches projects scaffolded manually by non-kit-aware agents (Copilot, plain Claude) that skipped `install.sh` . Bypass: `PSK_BOOTSTRAP_CHECK_DISABLED=1` . Kit-self auto-detected — skips symlink/config checks for the kit repo.
- **v0.6.5 — Register format improvements:** `reflex/lib/update-eval-trace.sh` parses cycle from pass-name ( `cycle-NN-pass-NNN` ) when `.cycle-meta` has stale `cycle=0` , adds `coverage.md` drill-down, extracts "Tests exercised:" summary line per pass — surfaces PASS-side data, not just failures.
- **v0.6.6 — Reflex Phase 0 pre-compute (claim-driven probing + reference-state diff + assumption surfacing):** closes the meta-gap surfaced in v0.6.5 review where reflex QA ran 3 GRANTED passes on an infrastructurally-empty project because the probe set was a closed 16-dim checklist. Moves reflex from "fixed checklist" to "probe-set derived from the project itself," while keeping the 16-dim checklist as a safety net. **Mode A** ( `reflex/lib/extract-claims.sh` ): walks docs and extracts every claim with `probe_type` + `probe_target` . **Mode B** ( `reflex/lib/state-diff.sh` + `reflex/reference-state/speckit-project.yaml` ): reference definition of complete speckit project; deltas pre-classified by severity; CRITICAL auto-promote to findings. **Mode C** (in `qa-agent.md` ): every pass writes `assumptions.md` ; unverifiable assumption = MAJOR finding. Both helpers run in `spawn-qa.sh` BEFORE sandbox creation. Dev-Agent expanded: fix findings + build unfulfilled claims + remediate state-diff deltas.

Deterministic bash (<1s, <100 tokens pre-compute) saves per-pass LLM budget for semantic verification.

- **v0.6.7 — Reflex 7-layer Senior/Principal-level QA system:** the user must NOT be QA of their own project. Reflex now operates with a 7-layer Senior/Principal-level QA system that derives all probes from the project's spec pipeline + kit toolkit + running system — zero human-curated lists. **Layer 1** ( `reflex/lib/check-rft-integrity.sh` ): R→F→T pipeline integrity per `[x]` feature (Rn map + criteria block + design plan + non-trivial test + TASKS mark). **Layer 2** ( `reflex/lib/doc-code-diff.sh` ): bidirectional doc↔code consistency. **Layers 3-5/7:** qa-agent.md mandates (behavioral 1+5+3+1 per feature / test-quality audit / cross-feature integration / external CVE+OWASP+deprecation research). **Layer 6:** spawn-qa.sh captures `psk-sync-check` , `psk-doc-sync` , `test-release-check` , `psk-code-review` outputs as research input. Senior/Principal-level philosophies codified in qa-agent.md + dev-agent.md prompts. 16-dim checklist preserved as safety net; CRITICAL signals from Layer 1/6 short-circuit it. Architecture plan committed at `agent/design/f70-reflex-senior-engineer-qa.md` . Companion Dev plan at `agent/design/f70-reflex-senior-engineer-dev.md` . Layer 4 hardening log shipped ( `reflex/history/hardening-log.md` + `reflex/lib/log-hardening.sh` ).
- **v0.6.8 — Reflex 7-layer full implementation: Layer 3/5/7 QA seed-helpers + Layer 7 Dev ADL extraction:** completes the v0.6.7 architecture by shipping deterministic helpers for Layers 3, 5, 7 (QA) and Layer 7 (Dev). All 7 layers now have deterministic-bash seed-helpers wherever possible — LLM creative work goes to genuine reasoning. **scaffold-behavioral-tests.sh** generates 1+5+3+1 test-plan skeletons per `[x]` feature with criteria pulled from `### F{N}` blocks. **identify-integration-probes.sh** finds feature-pair candidates from design-plan + criteria-block cross-references. **external-research.sh** scans manifests (package.json / requirements.txt / go.mod / Dockerfile) and seeds CVE / framework-changelog / OWASP queries; includes stack-specific recent-vuln hints (e.g. prompt injection per OWASP LLM Top 10). **auto-extract-adl.sh** drafts ADL entries from Dev's commit-body rationale post-pass. `spawn-qa.sh` Phase 0 produces 8 pre-populated artifacts. Field-tested on kit-self: 70 test-plan skeletons in <1s, 77 candidate pairs found.
- **v0.6.9 — The kit is audited and improved by the reflex it built:** the architecture built across v0.6.5-v0.6.8 was designed to find drift in user projects. v0.6.9 turned that exact machinery against the kit itself. The same QA-Agent that catches bugs in your code caught real bugs in the code that builds the QA-Agent — Phase 0 exposed 180 mechanical drift items (170 R→F→T pipeline breaks + 10 doc↔code drifts) the kit's own existing mechanical gates had passed clean for months; QA filed 5 findings; Dev closed 2 inline (live `kit-loop-state.yml` → `loop-state.yml` rename incompleteness from v0.6.2 + Layer-4 hardening shipping new `PSK018` per-feature pairwise check that would have caught 69-of-70-criteria-blocks-deleted under the existing shallow gate). Layer 5 hub-suppression added (77→13 pairs after F64+F70 hubs filtered). Field-test artifacts committed in `reflex/history/field-test-v0.6.8/` as permanent evidence.

*A tool sufficient to audit user projects must be sufficient to audit the project that built it*  
— v0.6.9 proved that proposition.

- **v0.6.38 — Kit-quality fixes: playwright gate, verdict integrity, findings.yaml discipline, flow doc coverage (May 2026):** 9 targeted kit fixes from searchsocialtruth cycle-07 kit-scope findings (kit-first philosophy). **Playwright gate fix ( gates.sh ):** `npm ci --prefer-offline --silent` runs before `npx playwright test`; `playwright_suite_gate_enabled` config key added. **write\_verdict() structural field ( run.sh ):** `verdict.md` now emits `verdict: GRANTED|DENIED` as explicit field for convergence-audit gate. **findings.yaml status discipline ( dev-agent.md ):** Dev-Agent Step 4 mandates updating `status: open → closed`; `[source: <pass-name>]` promoted to HARD REQUIREMENT. **track-tokens.sh unbound variable fix. Kit version drift advisory ( psk-sync-check.sh ):** new `check_kit_version_drift()` compares project `Kit:` field against installed version. **.gitignore playwright output:** `test-results/` + `playwright-report/` added. **REQUIREMENTS.md duplicate advisory ( mandate-audit.sh ):** ADVISORY when both root `REQUIREMENTS.md` and `agent/REQS.md` exist. **Feature-named test files mandate ( psk-orchestrate.sh ):** Phase 7 requires `tests/f{N}-feature-name.test.ts` naming. **REQUIREMENTS.md deleted from kit root — agent/REQS.md** is authoritative. **Flow doc coverage (13 + 17):** 13-release-workflow.md Step 6 + 17-reflex.md §v0.6.30/§v0.6.35 sections. 1982 tests (1837 framework + 145 benchmarking).
- **v0.6.37 — Loop Iteration 8: QA orchestration (wave-based parallel dim-agents) + Dev-Agent Phase 1 analysis (May 2026):** solves the kit-self reflex SDK stream-idle-timeout (G-KIT-QA-PROMPT-LENGTH-01). New `reflex/prompts/qa-agent-orchestrator.md` (orchestrator: Phase 0 project understanding + wave planning + parallel dim-agent spawning + aggregation/de-duplication) + `reflex/prompts/qa-agent-dim.md` (focused dim-agent: receives Phase 0 summary + assigned dim slice). `spawn-qa.sh` auto-detects orchestrator/dim prompts → uses `qa_mode: orchestrated` when both present, falls back to monolithic. `reflex/config.yml` new `qa_agent:` section: `max_dims_per_spawn: 10` + `max_parallel_agents: 4`. Dev-Agent Phase 1 analysis: read-only root-cause grouping → `fix-plan.md` checkpoint → Phase 2 cascade-check execution. `reflex/config.yml` new `dev_agent:` section: `strategy: single_pass_ordered`, `analysis_phase: enabled`, `cascade_check: after_each_commit`. 1982 tests (1837 framework + 145 benchmarking).
- **v0.6.36 — Loop Iteration 7 kit-side: in-reflex deferrals closed (P/Q/R/S) + machinery propagation (May 2026):** closes all 4 in-reflex deferred items from v0.6.35. Phase P: `dev-self-verify.sh` flipped from opt-in to default-on (10th gate active in every reflex run). Phase Q: `psk-version-cascade.sh` extended for kit-machinery propagation via `.portable-spec-kit/sync-targets.txt`. Phase R: dev-self-verify smart DSL expanded (3→8+ patterns: contains, matches, is exactly, exit code N, has N+ lines, file exists). Phase S: `EXPECT_RESUME` pattern generalized in `loop.sh::_l2_iter_trap`. 10 new assertions in `tests/sections/04-reflex.sh`. 1955 tests (1810 framework + 145 benchmarking).

- **v0.6.35 — Loop Iteration 6.5 kit-side: convergence-rule fix — no budget-stops, set-based plateau (May 2026):** closes the budget-driven leftover-deferral pattern. Phase G: `convergence:` config rewrite — only 4 STOP signals (GRANTED, REGRESSION, set-based PLATEAU, infinite\_loop\_protection:100); `budget:` section REMOVED, replaced with `guidance:` (recommended targets, never hard caps). Phase H: set-based PLATEAU detection — compares finding ID sets, stops only on EXACT match. Phase I: strip budget caps from sub-agent prompts. Phase J: cost reporting first-class (QA tokens · calls · USD · Dev tokens · calls · USD per pass + cumulative). 1957 tests (1805 framework + 145 benchmarking).
- **v0.6.34 — Loop Iteration 6 kit-side: structural recurrence prevention (Phase A-F) + version-cascade ceremony (May 2026):** closes 3-iteration sub-agent claim recurrence pattern. Phase A: `reflex/lib/dev-self-verify.sh` — 10th mechanical gate (REFLEX\_DEV\_SELF\_VERIFY). Phase B: regression tests pin RFT/AB/DCD fixes in `tests/sections/04-reflex.sh`. Phase C: `agent/scripts/psk-version-cascade.sh` (script #27) — field-anchored version propagation. Phase D: `clear_gate_caches()` in `gates.sh` — wipes `agent/.release-state/` cache before each gate iteration. Phase E: L2 EXIT trap honors EXPECT\_RESUME=1. Phase F: resume-qa/resume-dev auto-mirror result files. 1933 tests (1788 framework + 145 benchmarking).
- **v0.6.33 — Loop Iteration 5 kit-side: 4 deep fixes from v0.6.32 backlog + cycle-06 dev-fixes (May 2026):** L5.1 (1e3f607): RFT comma-split fix in `check-rft-integrity.sh`. L5.2 (0ea56d2): abort-integrity schema v2 split (abort\_findings vs baseline\_audit\_trail BASELINE tier). L5.3 (8951bf7 + 311b607): doc-code-diff comprehensive kit-installed-path exclusion. L5.4 (5205f52): pre/post-clean state in `03-reliability.sh` + `05-mandate-compliance.sh`. V5.1 (035eab9): README badge v0.6.32→v0.6.33. V5.2 (e62b25d): prep release framework version bump. doc 19 §15: Iteration 5 lessons — PLATEAU rule clarified. Project searchsocialtruth v0.4.7 reflex cycle-09 GRANTED all-9-gates-green. 1909 tests (1764 framework + 145 benchmarking).
- **v0.6.32 — Loop Iteration 4 kit-side: Per-feature test architecture (Approach 3) + 4 deferred fixes cleared (May 2026):** architectural fix for the cost wall surfaced by Loop 3 G-KIT-PHASE0-COST-01. Each of 70 kit features now has a dedicated `tests/features/fNN-*.sh` selective audit (~1 sec each) instead of pointing at the 1909-test monolith. Release-check runtime: hours → 15.8 seconds. ADR-077 (T1.1: playwright gate ulimit fix) · ADR-078 (T1.2: RFT comma-split parity completion) · ADR-079 (T1.3: abort-integrity BASELINE tier) · ADR-080 (T1.4: doc-code-diff kit-script exclusion) · ADR-081 (T2: tests/{shared,features}/ skeleton + dynamic discovery) · ADR-083 (T4: 6 shared helpers) · ADR-084 (T5-T7: 70 per-feature audit files) · ADR-085 (T8: SPECS.md Tests column migrated) · ADR-086 (T9: sections/\* deprecation headers). Sections retained as exhaustive backbone. 1909 tests (1764 framework + 145 benchmarking).
- **v0.6.31 — Loop Iteration 3 kit-side: Convergence-discipline structural enforcement + 5 deferred fixes cleared (May 2026):** convergence-discipline elevated from documentation to gate-enforced invariant. ADR-066 (L1: abort-detection in preconditions + `--recover-from-abort` flag) · ADR-067 (L2: EXIT-trap completion

contract — INTERRUPTED verdict.md on any exit) · ADR-068 (L3: per-iteration `.iter-status.yml` audit trail) · ADR-069 (L4: abort-integrity probe in Phase 0 pre-compute) · ADR-070 (L5: 9th mechanical gate `convergence-audit` rejecting INTERRUPTED verdicts) · ADR-071 (L6: `portable-spec-kit.md` §Convergence section + 9-gate doc count refresh). Plus 5 v0.6.30 deferred fixes (M1-M5): ADR-072 subshell env-var propagation · ADR-073 Phase 0 cache sharing · ADR-074 RFT comma-split parity · ADR-075 PSK001 version-scoped grep · ADR-076 doc-code-diff kit-script exclusion on user projects. 1909 tests (1764 framework + 145 benchmarking).

- **v0.6.30 — Loop Iteration 2 kit-side: Mandate-Compliance probe (Dim 25) + 8th gate + 6 deferred findings cleared (May 2026):** adds the structural-completeness floor pairing with Dim 24's ceiling. F1 ( `agent/scripts/mandate-audit.sh` ): deterministic kit-mandate compliance probe with MAJOR/MINOR findings. F2 — QA-Agent Dim 25 registered: `reflex/prompts/qa-agent.md` header 24→25 dims + new Dimension 25 section (5 probes). F3 — 8th mechanical gate: `reflex/lib/gates.sh` registers `mandate-compliance` gate. F4 ( `tests/sections/05-mandate-compliance.sh` ): 9-test regression suite. F5 ( `portable-spec-kit.md` ): doc refresh. G1 — Phase 6.5 pre-flight: `reflex/lib/orchestration-phase-6-5.sh` runs mandate-audit before reflex. G2 — `psk-orchestrate.sh` : wires Phase 6.5 into orchestration pipeline. G3 — `docs/work-flows/20-orchestration-completeness.md` : flow doc. 6 deferred findings (H1 — QA-KIT-RUNNER-DETECT-01 · H2 — QA-KIT-PSK002-CROSSVER-01 · H3 — QA-KIT-INSTALLER-MANIFEST-01 · H4 — QA-KIT-RELEVANCE-NOISE-01 · H5 — QA-KIT-SELFTTEST-ISOLATION-01 · H6 — QA-META-PHIL-CYCLE-05-01). 1909 tests (1764 framework + 145 benchmarking).
- **v0.6.29 — User-project reflex correctness — 5 fixes G19-G23 from searchsocialtruth field-test:** first user-project field-test results land via the kit-evolution protocol. Reflex run on searchsocialtruth (v0.1.1) surfaced 5 kit-machinery bugs filed as `scope: kit` . **ADR-041 (G23):** kit-self detection uses kit-only directory discriminator (examples/starter + examples/my-app + tests/sections + install.sh + agent/PHILOSOPHY.md) — replaces broken symlink-vs-real-file heuristic. Cascade fix unblocking correct routing for every other QA-KIT-\* finding. **ADR-042 (G19):** sandbox cp-fallback enumerates dynamically — catches `src/` , `app/` , `components/` , `packages/` on Next.js / monorepo projects. **ADR-043 (G20):** rft-integrity regex matches h2-h6 R-rows. **ADR-044 (G21):** release-check `check_test_relevance()` advisory check. **ADR-045 (G22):** doc-code-diff `IS_KIT_SELF` skip clause for kit-installed paths on user projects. 1756 tests passing (1611 framework + 145 benchmarking); 9 N84 regression sub-tests cover all 5 fixes. [source: `avacr-eval-searchsocialtruth/cycle-01/pass-001`]
- **v0.6.27 — Symmetric self-evolution + Tier 3 auto-probe-synthesis + bookend release ceremony + finding-closure helper + cycle-id rule relaxation:** major kit-evolution release closing the v0.6.7+ residual auto-evolving-QA backlog plus a structural blind spot the user surfaced after Mode C iteration (kit was biased toward addition and blind to consolidation). **ADR-038 — Tier 3 auto-probe-synthesis:** `agent/scripts/psk-blind-spot-synthesize.sh` (~280 LOC) reads `qa-blind-spots.md` open entries,

classifies target into 5 buckets, scaffolds PR-style proposals at `agent/tasks/proposed/Gxx-<slug>.md`. **ADR-039 — Reflex bookend release ceremony:** iter 1 = prep-release · iters 2+ skip ceremony entirely · GRANTED runs ONE final refresh-release. N-iteration cycle now runs ceremony exactly 2× (was N×). **ADR-040 — Symmetric Self-Evolution (P9):** 5 layers — `agent/PHILOSOPHY.md` P9 principle · `psk-coverage-overlap-check.sh` (~210 LOC) · `/optimize` cat 14 (probe redundancy) · Phase 6 evolution-gauntlet Gate G · QA Dim 24 (3 probes filing `QA-OVERLAP-NN`) · Phase 5 mandate ≥1 audit-coverage OVERLAP observation · `OL-NNN` registry parallel to `BS-NNN` · `psk-close-finding.sh` closes findings without YAML hand-edit. Cycle-id rule relaxed to verdict-agnostic (closes DENIED-then-externally-fixed trap). Kit bug fix: `psk-bootstrap-check.sh` C7 kit-self detection. Backlog cleanup 48→17. 1879 tests (1734 framework + 145 benchmarking); 22/20 sync-check; new N80 (12) + N81 (6) + N82 (15) regressions plus 8 cycle-04 regressions. *Closes auto-evolving-QA loop; kit now self-detects probe redundancy alongside probe gaps.*

- **v0.6.26 — Per-cycle pass numbering + summary.csv schema v5 + reflex self-test convergence:** closes user-reported "cycle 2 showing pass 3,4,5 instead of 1,2,3" bug. Pass numbers reset per cycle (each cycle starts fresh at `pass-001`); composite key `(cycle, pass)` gives unique row identity in `summary.csv` (schema v5 adds `cycle` column). Lifts ADR-037 (reflex self-test 8-finding closure landing in v0.6.25 patches) into a formal release.
- **v0.6.25 — Post-merge soak gauntlet + Tier 1/4/5 auto-evolving QA:** ADR-035 ships `agent/scripts/psk-soak-schedule.sh` (Phase 6 (e) post-merge 48h soak, deferred from v0.6.24). ADR-036 closes Tier 1 (BS-NNN registry seeding), Tier 4 (auto-promote OPEN → CLOSED on Gxx merge), Tier 5 (count-based gauntlet probes for blind-spot density tracking). ADR-037 (reflex self-test 8-finding closure landing in patch series).
- **v0.6.24 — Self-evolving-plan gap-closure (hygiene release):** three rounds of audit-and-close on the v0.6.16-23 plan that surfaced 9 sub-deliverables headline commits had claimed-shipped but were missing. Round 1 closed 7 phase artifacts. Round 2 closed `install.sh` gaps. Round 3 closed 5 flow-doc omissions. Phase 6 (e) deferred to v0.6.25. *Pure hygiene; lesson: companion artifacts must commit alongside headline deliverables.*
- **v0.6.13 — Nested per-cycle history layout + Dim 23 + cycle-summary aggregator + autoloop convergence:** closes the meta-gap class user surfaced in cycle-01 trace audit (5 issues + 4 sub-gaps). Formalizes **Dim 23 — Auditor-output hygiene** with five new probes (register hygiene · cost data persistence · parallel-directory disambiguation · cross-surface terminology consistency · self-output validation loop) so QA self-evolves to catch this class going forward. Migrates reflex history from flat `cycle-NN-pass-NNN/` to nested `cycle-NN/pass-NNN/`; sandbox mirrors. Per-cycle parent dir hosts a `summary.md` aggregator (`reflex/lib/cycle-summary.sh`). Drops the hard 3-iter autoloop cap in favor of convergence-based stopping (GRANTED · REGRESSION · findings\_floor · plateau · fix-rate drop); `max_iterations_safety` default 20 is escape hatch only. First successful Dev-Agent run end-to-end: cycle-01-

pass-004 closed QA-NOISE-RE-01 bucket A, all 6 mechanical gates green, fast-forward auto-merge. 3-layer QA→Dev contract enforcement closes "Dev never spawned on partial QA timeout" pattern. *Auditor-output hygiene formalized; reflex history reshaped per-cycle; autoloop runs to convergence.*

- **v0.6.12 — Reflex iter-1 convergence anchor + close all 15 v0.6.11 audit findings:** anchor version for the reflex iter-1 convergence cycle. Consolidates 8 refresh-release commits from v0.6.11 baseline. All 15 audit findings (5 pre-existing ARBs + 10 iter-1 QA findings) resolved or properly deferred. Major perf win: persistent test-ref cache cold 54s → warm 5s = 10.9× speedup. Dim 17 cost/perf audit + Dim 18 philosophy self-audit added to QA prompt + Dev mirror principles 8/9. pass\_score (Dim 17 ranking) v3 summary.csv schema. Option C thematic test-suite split: `tests/lib.sh` + `tests/sections/{01-infrastructure,02-pipeline,03-reliability,04-reflex}.sh` + thin orchestrator. *Iter-1 self-audit complete — kit closed every finding it surfaced about itself.*
- **v0.6.11 — Reset allowlist + iter-aware refresh-release optimization:** two kit-infrastructure improvements driven by usage friction during v0.6.10 wrap-up. (1) `reflex/lib/reset.sh` switched from hardcoded glob lists to allowlist-based `find` traversal with `HISTORY_KEEP=("hardening-log.md")`. Symmetric `--reset-hardening` flag for true clean-slate. (2) `reflex/lib/loop.sh` Phase 1 now branches on `ITER : iter 1 = prepare release` (version bump), `iter 2+ = refresh release` (no bump — captures Dev's auto-merged fixes through same critic gates at ~50% cost). +7 new test assertions. Doc-coverage backfill closes 24 cumulative `psk-doc-sync.sh` gaps from v0.6.5-v0.6.10.
- **v0.6.10 — Closing v0.6.9 ARBs: kit's own RFT debt backfilled + PSK018 strict by default:** acted on the v0.6.9 field-test verdict. Backfilled all 55 missing `### F{N}` acceptance criteria blocks in `agent/SPECS.md` (F1-F55, 3-5 testable bullets each, ~200 lines), closing QA-PIPE-FT-01-ARB. Flipped PSK018 (per-feature pairwise check in `psk-sync-check.sh`) advisory → strict-by-default; future commits adding an `[x]` feature without matching block fail pre-commit. Opt-out: `PSK_FEATURE_CRITERIA_STRICT=0`. Layer 1 RFT integrity on kit-self: 55 C2-criteria breaks → 0.
- **14 kit framework gaps closed (G1-G15 less G11):** parser + regex consistency across `reflex/lib/` (G1+G2+G5+G6); `test-templates.md` skill with live-uvicorn fixture (G7); G11 pre-shipped via canonical trace template; rate-limit escape-hatch docs (G13); serial-execution trade-off disclosure in `f70-reflex` design (G14); `spawn-qa` SQLite artifact purge (G9); per-pass token accounting with v2 summary.csv schema (G15); `file-bugs.sh` schema validator + dim-floor enforcement (G8); machine-readable coverage YAML block (G10); minimum-per-dimension probe count rule (G12); preconditions accept reflex-install commits (G3); `--recover` diagnostic plus `reflex/lib/recover.sh` for partial Dev state (G4).
- **MINIMAL template (3rd tier) for short answers:** breadcrumb + border + headline + optional one-paragraph body + arrow. Target ~2s gen time, ~80-100 tokens. Use for

yes/no, status checks, short factual recalls, simple acknowledgments. BRIEF remains default for multi-dimensional answers; DETAILED on explicit depth request.

- **Auto-dispatch rule:** trigger table maps answer shape to template tier — agent picks, user does not specify. Override phrases — "shorter" drops a tier, "more depth" promotes one, "raw reply" strips scaffolding.
- **Writing Style sub-section (5 editorial rules):** one idea per sentence · periods over em-dashes · drop semicolons · cut parenthetical asides unless load-bearing · one voice per reply.
- **Breadcrumb rules 6a + 6c + 6d:** trailing ✓ -hint of most-recently-closed sibling (6a); skip breadcrumb on consecutive MINIMAL replies in same node (6c); arrow footer optional in MINIMAL with no next action (6d). Rule 6b (IDs-only compact) shipped then reverted per user feedback — clarity beat the ~15-token savings.
- **Pre-send self-check:** mandatory 4-anchor check (breadcrumb, border, template body, arrow footer) run silently before every reply. No exemptions for "let's discuss" or "status update" or post-compact replies.
- **3-tier long-session scalability:** T1 auto-collapse 3+ consecutive ✓ siblings into `Nx-Ny ✓ <summary>` ; T2 default-render active chain on "where was I" (full tree on "show full stack" ); T3 mid-session chapter rotation at ≥15 KB or ≥80 nodes into `.session-archive/YYYY-MM-DD-chapter-NN.md` .
- **Promotion rule:** closed ✓ session-stack node with durable project work (code/docs changes) prompts for `agent/TASKS.md` entry before session end.
- **Pre-session read list updated:** `.session-stack.md` added to the session-start read order (item 4 of 11).
- **Skill trigger for "where was I" / "losing context" / "conversation-stack":** auto-loads `.portable-spec-kit/skills/session-trace.md` .
- **Skill trigger for "integration tests" / "FastAPI" / "live-server":** auto-loads the new `.portable-spec-kit/skills/test-templates.md` .
- **Rule-revision post-mortem meta-rule:** an ADL entry is required in `agent/PLANS.md` when any framework rule is revised ≥3 times in a single session (learning-from-churn).
- **Unpushed-commit advisory in psk-sync-check:** `--full` mode warns at ≥10 unpushed commits on current branch. Threshold configurable via `PSK_UNPUSHED_WARN` .
- **Signoff discipline:** `signoff.md` lists blocking findings explicitly; verdict is GRANTED only when zero CRITICAL + zero MAJOR blocking. Kit maintainer routes `scope: kit / meta` findings to `agent/tasks/Gxx-*.md` via `file-bugs.sh` .
- **Prompts rewritten:** QA-Agent + Dev-Agent prompts updated for adversarial framing, sandbox mechanism reference, citable-quote gate, and trace continuity across passes.
- **Paper title update:** "Adversarial Verbal Actor-Critic Refinement (AVACR)" — see v0.6.0 bullet above for full title and disambiguation.
- **SPD paper v2 §7.6 cross-reference:** paper methodology section now cross-references AVACR evaluation harness.

- **Design plan §4-5 and §18:** `agent/design/f70-reflex.md` §4 (QA-Agent), §5 (Dev-Agent), §18 (serial execution trade-off) updated for v0.6.0 shape.
- **Adversarial-goal reframing:** see v0.6.0 bullet above — this v0.6.0 delta narrows the framing to explicit convergence criterion.
- **Sandbox mechanism:** see v0.6.0 bullet above — v0.6.0 adds dual-rooted access reference documentation.

## v0.5 — Full Pipeline + Reliability Architecture (April 2026)

Kit: v0.5.1 — v0.5.23

- **Full development pipeline (F63–F69):** REQS → SPECS → PLANS → DESIGN → TASKS → RELEASES with `RESEARCH.md` support. Jira integration ( `psk-jira-sync.sh` ), auto code review ( `psk-code-review.sh` ), scope drift detection ( `psk-scope-check.sh` ), project config, onboarding tour, kit self-help, requirements + research pipeline, skill-based architecture.
- **v0.5.8 — Reliability Architecture (3-layer enforcement):**
  - **Layer 2A — Bash critic:** `psk-sync-check.sh` runs 18 deterministic checks (version consistency, test count, flow doc count, feature count, R→F→T gate, script perms, required dirs, CHANGELOG/RELEASES version, ARD content, AGENT.md stack, README Latest Release, README install list, README agent table, README flow table, flow doc content, critic prompts meta-check, secrets scanning). Mode auto-detection (kit/node/python/go/rust/generic/custom). `--quick` (<500ms, version+tests only) and `--full` (all 18 checks) modes. Structured error codes PSK001–PSK017. Silent on clean. Sensitive data exclusion ( `.env` , `*.pem` , `*.key` , `credentials.*` ). Emergency bypass via `PSK_SYNC_CHECK_DISABLED=1` .
  - **Layer 2B — Sub-agent critic:** `psk-critic-spawn.sh` file-based protocol ( `critic-task.md` + `critic-result.md` ). Three templates: flow-docs verification, release-notes verification, final-validation comprehensive sweep. 5-iteration cap with escalation. CURRENT/STALE structured output.
  - **Layer 3 — Hooks:** `.claude/settings.json` with PostToolUse hook (warns on edit drift, silent on clean). `.git/hooks/pre-commit` blocks commits that fail sync-check. Hooks run outside agent control — agent cannot bypass. `psk-install-hooks.sh` installer detects Husky/pre-commit framework, wraps existing hooks without overwriting.
- **Release pipeline rewritten ( `psk-release.sh` ): 10 steps with artifact gates. Validation moved to Step 9 (final gate) from Step 4. Step 6 (Version Bump) fully automated — reads current version from `AGENT_CONTEXT.md` , computes next patch, seds all files, verifies via sync-check. Steps 4 and 8 agent-required with artifact verification (mtime + content checks). State persisted in `agent/.release-state/` .**
- **End-user distribution ( `install.sh` ): One-command install: `curl -fsSL .../install.sh | bash` . Downloads 12 scripts + 17 skill files + templates. Auditable (prints summary before execution). `--yes` for non-interactive. `--from` for local testing. Symlinks for all agents (`CLAUDE.md`, `.cursorsrules`, `.windsurfrules`, `.clinerules`, `.github/`**

copilot-instructions.md). Clean uninstall via `psk-uninstall.sh` (preserves user pipeline files).

- **Context reduction:** Core framework file 1998 → 912 lines (54% reduction). 894 lines extracted to 17 skill files loaded JIT at peak attention (mitigates "Lost in the Middle" attention gradient). New skill files: `hooks-and-critics.md`, `self-help.md`, `ci-setup.md`, `config-details.md`, `init-process.md`, `project-setup.md`, `onboarding-tour.md`, `dashboard.md`, `multi-agent.md`, `jira-integration.md` (plus existing templates, python-environment, source-structures, profile-setup, document-generation, test-release-check-template, release-process).
- **Research document:**  
`agent/research/plans/r1-agent-reliability-architecture.md` (3600 lines, 24 sections). Synthesizes 5 academic papers (Lost in the Middle, ReAct, Process Reward Models, Reflexion, Self-Consistency), production framework analysis (LangGraph, Temporal, OpenClaw), CI/CD patterns (Make, Bazel, Terraform), and Google Research multi-agent critic work (PaperVizAgent, ScholarPeer). Documents token optimization strategy (architecture is net token-negative vs baseline when optimized).
- **v0.5.21–v0.5.23 — F70 Reflex (Adversarial Verbal Actor-Critic Refinement Loop, AVACR):** Post-prep-release automated adversarial QA + auto-fix loop with asymmetric goals (QA fails / Dev foolproofs). Fresh-context sandboxed *QA-Agent* (Critic, senior-engineer auditor, 14-dimension hunt, 4-persona testing, research-grounded findings with citable-quote honesty gate, `AGENT.md` + `AGENT_CONTEXT.md` excluded from sandbox worktree) reads the full speckit pipeline (REQS/SPECS/PLANS/DESIGN/RELEASES + docs + running system) and adversarially exercises every SPECS acceptance criterion + CHANGELOG claim through the project's real public surface. Files peer-exchange `findings.yaml` with priority + blast-radius + confidence calibration. Fresh-context *Dev-Agent* (Actor, peer fixer with adversarial foolproof goal, 4-bucket diagnosis + sibling-class hardening + Bucket-D grounded rebuttals + human-arbitration routing) fixes atomically with per-commit mechanical gates, 1 finding = 1 commit, max 3 retries before `[~]` escalation. Pass-over-pass regression-vector re-execution catches surface-patch fixes. `reflex/history/summary.csv` tracks surprise\_density trend. Precondition: HEAD must be a prep-release commit. Counters the "lost in the middle" failure mode where trivial Dev-written tests pass against stub implementations. `reflex/install-into-project.sh` deploys machinery into any speckit project with auto-detected gates. Section 60m adds 45 tests. First pass on the kit surfaced 5 real gaps existing structural gates couldn't see; Dev-Agent auto-fixed 7 of 8 findings with atomic commits. Docs: `docs/work-flows/17-reflex.md`; design plan `agent/design/f70-reflex.md`.
- **v0.5.23 — Full-Surface Documentation Coverage Analyzer ( `psk-doc-sync.sh` ):** CHANGELOG → full doc-surface coverage tool that extracts every feature from the current-minor CHANGELOG entry and verifies it's reflected across all 5 documentation surfaces: `agent/*.md`, `docs/work-flows/*.md`, `docs/research/*.md`, `ard/*.html`, `README.md`. Classifies each feature COVERED (≥2 surfaces) / PARTIAL (1 surface) / MISSING (0 surfaces) with surface-legend ``[AFPDR]`` and heuristic-suggested target doc per gap. Mandatory at release Step 4; `STEP_4_FLOW_DOCS` and `STEP_9_VALIDATION` critic

prompts wired to treat every MISSING line as STALE. `--strict` flag returns exit 1 for CI use. Section 60k adds 19 tests. First run surfaced 27+ MISSING features from v0.5.x as backlog for v0.5.23+ — the gate is now in place to prevent regression while that backlog is closed.

- **v0.5.10 — Dual-Gate Reliability Across All Executable Workflows:** Generic `psk-validate.sh <workflow>` helper terminates every workflow (release, feature-complete, init, reinit, new-setup, existing-setup) with the same bash + sub-agent pair. 5 new critic templates: `FEATURE_COMPLETE` (R→F→T + ADL), `INIT` (all 9 agent/\* populated from code), `REINIT` (no content loss during re-sync), `NEW_SETUP` (scaffold coherence), `EXISTING_SETUP` (non-destructive adoption). `psk-release.sh` Step 9 now delegates to `psk-validate.sh release`. Framework MANDATORY rule: agent runs the helper at end of every executable workflow. Section 60 (25 tests) verifies the wiring end-to-end.
- **v0.5.12 — Confidence Hardening (7 items):** Step 4 of release upgraded from mtime-only to per-flow-doc critic verdict ( `STEP_4_FLOW_DOCS` template required; every flow doc must have explicit `CURRENT:` by filename, missing coverage flagged, any `STALE:` blocks). Bypass audit log at `agent/.bypass-log` records every `PSK_SYNC_CHECK_DISABLED=1` or `PSK_CRITIC_DISABLED=1` use (timestamp, user, workflow); `sync-check --full` surfaces warning for any in last 24h. 5 workflow orchestrators ( `psk-feature-complete.sh` , `psk-init.sh` , `psk-reinit.sh` , `psk-new-setup.sh` , `psk-existing-setup.sh` ) add workflow-specific preflight: feature-complete enforces R→F→T + design plan + no TODO stubs; reinit snapshots agent/\* byte counts to detect content loss; existing-setup snapshots non-kit file sizes to detect destructive edits.
- **v0.5.13 — Secret Scanning (PSK011):** `check_secrets` added as 12th deterministic check in `psk-sync-check.sh --full`. Scans tracked files via single `git grep` for 12 high-signal credential patterns: AWS access/secret keys, GitHub PAT/fine-grained/OAuth/App tokens, Anthropic, Google, Slack, Stripe live/restricted, private key PEM headers. Placeholder-aware ( `paste-your` , `example.com` , `placeholder` , etc. excluded) + path exclusions ( `.env.example` , `node_modules/` , `tests/fixtures/` , binaries). Masked output prevents leakage to logs. PreCommit hook now blocks any commit containing tracked secrets — closes the gap between framework security rule and mechanical enforcement. Section 60d adds 11 tests.
- **v0.5.15 — Verbatim-Quote Critic (sub-agent honesty gate):** Closes the architectural ceiling on sub-agent honesty. Every `CURRENT:` verdict in `critic-result.md` now requires a `QUOTE:` line with a verbatim string (≥20 chars) from the named file — bash `grep -F` verifies the quote actually exists. Fabricated or missing quotes exit 3 and block the workflow. All 6 critic templates updated. Sub-agent honesty: ~70% → ~85%. Overall confidence: ~96%.
- **1222 tests** (1077 framework across 74 sections + 145 benchmarking). Section 59: Reliability Architecture (20 tests). Section 60: Dual-Gate Validation + Behavioral (28 tests). Section 60c: Workflow Orchestrators + Step 4 + Bypass Audit (21 tests). Section 60d: Secret Scanning (11 tests). Section 60e: Distribution Completeness (7 tests). Section 60f: Verbatim-Quote Critic (12 tests).

## v0.4 — CI/CD Pipeline + Spec-Based Test Generation (April 2026)

Kit: v0.4.1 — v0.4.11

- **GitHub Actions CI:** `ci.yml` runs all 3 test suites on every push and PR. `release.yml` verifies tag matches framework version on v\* tag push.
- **Framework CI/CD rules:** `ci.yml` template with stack-aware commands (Jest/pytest/Go/Bash), New Project Setup Step 7.5, Existing Project Setup checklist CI item, CI & Community Contributions section (4 rules).
- **Spec-based test generation (8 rules):** SPECS origin detection (forward vs retroactive), per-feature `### F{n}` acceptance criteria format, stub generation trigger, stack-aware stub formats, stub completion gate.
- **check\_stub\_complete():** added to `tests/test-release-check.sh` (kit copy + template) — blocks release if test stubs have unfilled TODO markers.
- **Community files:** PR template, bug report + feature request issue templates. CI badge in README.
- **Cross-platform sed fix:** `test-spd-benchmarking.sh` now runs on Ubuntu (GitHub Actions).
- **746 tests** (601 framework across 48 sections + 145 benchmarking). Section 42: CI/CD (29 tests). Section 43: Spec-Based Test Generation (15 tests).
- **v0.4.5:** 2 new flow docs (11-existing-project-setup, 12-cicd-setup). All 12 flow docs validated at 63-char box alignment. Section 19 expanded to cover all 12 flows. ARD flow diagrams section added. Prepare release rule updated: flow docs first, mandatory ARD audit, mandatory PDFs.
- **v0.4.6:** Flow docs reordered logically (onboarding 01-06, session 07-10, development 11-12, release 13). New `05-project-init.md` — explicit `init / reinit` commands. Renamed: `12-project-lifecycle.md`, `13-release-workflow.md`. `init / reinit` added as explicit command signals with full flows. Test Execution Flow extracted as named section — referenced by all test-running commands. Version bump BEFORE push rule added. 746 tests (601 framework + 145 benchmarking).
- **v0.4.7 — Team Intelligence Layer (F58–F62):** Progress Dashboard (trigger words → inline burndown with progress bars, BY CONTRIBUTOR for @username projects). Multi-Agent Task Tracking (@username ownership syntax, per-user task view, delegation/unassign protocol, cross-agent coordination via TASKS.md). Persistent Memory Architecture (named concept: 6 agent files = Persistent Memory, 5 properties: Durable/Shared/Portable/Team-scale/Auditable). Architecture Decision Log (ADL promoted to `## Architecture Decision Log` top-level in PLANS.md, ADR-NNN format, immutable history, supersede pattern). AI-Powered Onboarding (commit `agent/` for team/OS projects, clone → briefed, agent-agnostic). New flow doc: `14-team-collaboration.md`. Sections 44–48. 752 tests (607 framework across 48 sections + 145 benchmarking).
- **v0.4.8 — Release command fix:** `prepare release` corrected — steps 1–7 only, no commit, no push. Release summary shows GitHub/Tag as ⌚ pending. `prepare release` and `push` is the explicit command that commits and pushes. `13-release-workflow.md`

flow doc updated (7-step gate, corrected trigger table, 5 new edge cases). README commands table updated.

- **v0.4.9 — Release flow hardened + Kit Self-Validation + Portability Gate:** ARD audit rule changed to `ard/*.html` glob (no hardcoded filenames). WeasyPrint commands use loop form. Section 41 + 2 new tests: kit `agent/AGENT_CONTEXT.md` Version+Kit match; `CHANGELOG.md` built-over range includes current version. Section 41: 28→33 tests. ARD Guide version fix. Section 49: Kit Self-Validation (8 tests) — example symlink validity, Progress Summary, template compliance, root/Projects copy sync. Section 50: Framework Portability (11 tests) — no machine-specific paths, no hardcoded tool binary paths, WeasyPrint loop form, ARD glob, sync.sh/flow docs/ARD HTML clean. +7 symlink tests in Sections 6/7/11. 885 tests (740 framework across 58 sections + 145 benchmarking).

## v0.3 — Framework Hardening + R→F→T Traceability (April 2026)

Kit: v0.3.1 — v0.3.27

- **R→F→T traceability chain:** every done feature requires test reference in SPECS.md before release. `tests/test-release-check.sh` distributed as kit template, auto-created on project setup.
- **15 new enforcement sections:** SPECS.md staleness check, RELEASES.md trigger rule, scope change recording (DROP/ADD/MODIFY/REPLACE), no-slip task rule, pipeline sync, pre-release consistency gate (Section 41).
- **Release Process rule — full 8-step sequence:** tests → counts → version bump → PDFs → RELEASES.md → CHANGELOG.md → GitHub release → tag update. Explicit signals only.
- **Release command aliases:** `update release` = alias for `prepare release` (full 8-step); `refresh release` = re-test + sync current release without bumping version.
- **Release notes scope rule:** only include changes committed and visible in public repo; never mention excluded files (private docs, research papers).
- **Simplified versioning:** aligned patches (v0.N.x = release v0.N). All specific version examples replaced with generic `v0.N.x / v0.1.4→v0.1.5` so rules never go stale. GitHub release title = minor version (v0.N — Title) matching CHANGELOG.
- **README hardening:** Critical Scenarios table (8 situations), Development Guidelines section, "first methodology native to the AI era" framing.
- **746 tests** (601 framework across 48 sections + 145 benchmarking). All passing.

## v0.2 — Versioning + Project Scenarios (April 2026)

Kit: v0.2.1 — v0.2.9

- **Aligned versioning:** v0.N.x patches belong to release v0.N (e.g. v0.2.x = release v0.2). Auto-restructure agent files when framework version changes on pull.
- **Existing project setup:** guide don't force — scan, show checklist, user picks changes. Never rename/move/delete without approval.

- **9 project scenarios:** new, existing, monorepo, partial, cloned, subdir, multiple projects, returning, kit in one subdir only.
- **Project directory confirmation:** list directories, user picks or types new path before creating agent/ files.
- **Git rule:** check .git/ before committing — each project can be its own repo.
- **Version-based TASKS.md:** tasks grouped under release headings (v0.0 Done, v0.1 Done, v0.2 Done, v0.3 Current, Backlog).
- **RELEASES.md:** kit version range per release (v0.0.x for v0.0, v0.1.x for v0.1).
- **AGENT\_CONTEXT.md:** tracks Kit version — compared on session start to trigger restructure.
- **442 framework tests across 41 sections + 145 benchmarking tests (587 total).**  
**Added: Python/Conda env rules (27 tests), README validation (3 tests), requirements-to-delivery flow checks, 15 enforcement sections (staleness, RELEASES trigger, scope changes, no-slip, pipeline sync, security, push rules, git, existing projects, scenarios, retro fill, context persistence across 10 disruptions, R→F→T traceability chain).**

## v0.1 — Personalized Profile + Flow Documentation (April 2026)

Kit: v0.1.1 — v0.1.9

### User Profile

- **Profile moved to** `.portable-spec-kit/user-profile/user-profile-{username}.md`
- **Global home directory storage — asked once, works everywhere**
- **Per-user workspace files — committed, team-ready**
- **Username detection via** `git config user.name` **(slugified)**
- **First-time setup: GitHub fetch + 3 questions with RECOMMENDED defaults + Enter to skip**
- **New project: show profile, keep or customize, save to workspace**
- **Returning session: workspace exists → silent; only global → show/confirm**
- **12 edge cases handled (no gh CLI, empty files, multiple users, special chars)**
- **Cross-OS: macOS/Linux ( ~ ) + Windows ( %USERPROFILE% )**

### Flow Documentation

- **22 flow documents created in** `docs/work-flows/`
- **Each flow: title, trigger condition, step-by-step diagram, edge cases, files involved**
- **Context Management rule: update flows after implementation/tests**

## Framework

- 10 core strengths (added Personalized, Secure, restored Self-Scaffolding)
- 8 source code structure templates (added Mobile, Android Native, iOS Native, Full Stack + Mobile)
- Strict security rule — never expose API keys even if explicitly asked
- Context triad: context + flows + tests must always match code
- Guide moved from `docs/` to `ard/`
- Deleted `setup.sh` — agent handles profile creation

## Agent-Agnostic

- Removed all `.claude/` paths, `noreply@anthropic.com`, `Claude Opus`
- Generic `Co-Authored-By: AI Agent <noreply@ai-agent.dev>`
- 27 agent-switching tests verifying zero data loss

## Testing

- 242 tests across 22 sections (was 122)
- New sections: User Profile directory, first-time setup, new project, returning session, edge cases, flow documentation, ARD directory, context management

## v0.0 — Initial Release (March 2026)

- CLAUDE.md framework — portable spec-persistent development
- README with spec-kit comparison table
- CONTRIBUTING.md with PR guidelines
- PDF Quick Guide
- Two example projects (starter + my-app)
- GitHub repo created (aqibmumtaz/portable-spec-kit)
- Sync script for publishing
- 122 automated tests

## Table of Contents

---

1. Architecture Overview
2. User Profile System
3. Work Flows

- 4. Multi-Agent Support**
- 5. Security Model**
- 6. Source Code Templates**
- 7. Spec-Persistent Development Pipeline**
- 8. Testing Infrastructure**
- 9. File Structure & Purposes**
- 10. Deployment & Syncing**

# 1. Architecture Overview

The framework is a single markdown file that AI agents read as their instruction set. It creates a layered system:

```
Layer 1: portable-spec-kit.md           ← Framework (portable, shared)
Layer 2: .portable-spec-kit/           ← User profile (per-user, committed)
Layer 3: agent/AGENT.md                ← Project rules (per-project)
Layer 4: agent/AGENT_CONTEXT.md        ← Living state (per-session)
Layer 5: agent/SPECS → PLANNING → TASKS → TRACKER ← Dev pipeline
```

## Design Principles

| Principle          | Implementation                                                                          |
|--------------------|-----------------------------------------------------------------------------------------|
| Zero install       | One curl command downloads the file, symlinks for all agents                            |
| Personalized       | GitHub profile auto-detect, 3 preference questions, adapts behavior                     |
| Agent-agnostic     | Symlinks: CLAUDE.md, .cursorrules, .windsurfrules, .clinerules, copilot-instructions.md |
| Non-blocking       | Code first, specs retroactively — agent tracks silently                                 |
| Context-persistent | AGENT_CONTEXT.md updated every session, picks up after weeks                            |
| Self-validating    | Agent tests everything, fixes before presenting, 98%+ coverage target                   |
| Secure             | Never expose API keys — even if asked. Absolute rule, no exceptions.                    |

## File Hierarchy

```
workspace/
├─ portable-spec-kit.md           ← Framework (source)
├─ .portable-spec-kit/           ← Kit config
│   └─ user-profile/
│       └─ user-profile-{username}.md ← Per-user profile (committed)
```

[illegible]

## 2. User Profile System

### Purpose

**Tells the AI agent WHO it's working with — affecting response depth, technical language, analogies, and autonomy level.**

### Storage

| Location                                                                          | Purpose                      | Committed?           |
|-----------------------------------------------------------------------------------|------------------------------|----------------------|
| <code>~/.portable-spec-kit/user-profile/user-profile-{username}.md</code>         | <b>Global — asked once</b>   | <b>No (home dir)</b> |
| <code>workspace/.portable-spec-kit/user-profile/user-profile-{username}.md</code> | <b>Per-project, per-user</b> | <b>Yes</b>           |

### Lookup Order

- 1. Workspace** `.portable-spec-kit/user-profile/user-profile-{username}.md`
- 2. Global** `~/.portable-spec-kit/user-profile/user-profile-{username}.md`
- 3. Neither** → **first-time setup**

### Username Detection

`git config user.name` → **slugified (lowercase, spaces → dashes)**. Used for filename. `gh api user` **used only for greeting name/bio**.

### Profile Format

```
# User Profile
> Auto-created on first session. Edit anytime.

- **Name** — Education. Expertise.
- Communication style: {selected or custom}
- Working pattern: {selected or custom}
- AI delegation: {selected or custom}
```

## Preference Options

| Question        | Options                                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Communication   | (a) direct and concise ← RECOMMENDED<br>(b) direct, data-driven, comprehensive<br>(c) conversational and collaborative                     |
| Working pattern | (a) iterative — starts brief, expands ← RECOMMENDED<br>(b) plan-first — full specs before code<br>(c) prototype-fast — build quick, refine |
| AI delegation   | (a) AI does 70%, user guides 30% ← RECOMMENDED<br>(b) AI does 90%, user reviews 10%<br>(c) 50/50 collaboration                             |

## Edge Cases

| Scenario                 | Handling                            |
|--------------------------|-------------------------------------|
| No gh CLI                | Ask name/expertise manually         |
| GitHub name empty        | Use login as fallback               |
| GitHub bio empty         | Ask for education/expertise         |
| Empty profile file       | Treat as missing, run setup         |
| Multiple users (team)    | Each user has own file              |
| Special chars in name    | Slugified for filename              |
| Agent can't write        | Show content, user creates manually |
| User skips all questions | Recommended defaults applied        |

### 3. Work Flows

14 documented flows in `docs/work-flows/` covering every system interaction:

| #  | Flow                           | Trigger                             | Key Actions                                                         |
|----|--------------------------------|-------------------------------------|---------------------------------------------------------------------|
| 01 | First Session in New Workspace | New workspace                       | Environment detection → <b>WORKSPACE_CONTEXT</b> → auto-scan        |
| 02 | User Profile Setup             | First time using kit                | GitHub fetch → 3 questions → save global + workspace                |
| 03 | New Project Setup              | Creating new project                | Show profile → keep/customize → scaffold agent/ + dirs              |
| 04 | Existing Project Setup         | First-time kit onboarding           | Detect state → scan → fill agent/ → checklist → apply selected      |
| 05 | Project Init & Reinit          | <code>init / reinit</code> commands | Confirm dir → deep scan → create/update agent/ files → show summary |
| 06 | CI/CD Setup                    | New/existing project setup          | Stack detect → ci.yml template → badge → branch protection          |
| 07 | Returning Session              | Coming back to project              | Load profile → read context → summarize → continue                  |
| 08 | Agent Switching                | Changing AI agent                   | All agents read same file via symlinks → zero data loss             |
| 09 | Profile Customization          | Different prefs per project         | Show <b>CURRENT + RECOMMENDED</b> → save local override             |
| 10 | File Management                | Agent encounters managed file       | Create / restructure / leave-as-is — never lose content             |
| 11 | Spec-Persistent Development    | Any workflow — specs persist        | <b>SPECS → PLANS → TASKS → RELEASES</b> — living specs, never block |
| 12 | Project Lifecycle              |                                     |                                                                     |

|    |                    |                              |                                                                     |
|----|--------------------|------------------------------|---------------------------------------------------------------------|
|    |                    | Client requirements received | R→F traceability → scope changes → F→T coverage → handoff           |
| 13 | Release Workflow   | prepare release command      | Tests → counts → version bump → PDFs → publish → tag                |
| 14 | Team Collaboration | Multi-agent / team project   | @username tagging → per-user view → delegation → progress dashboard |

Context Management Rule

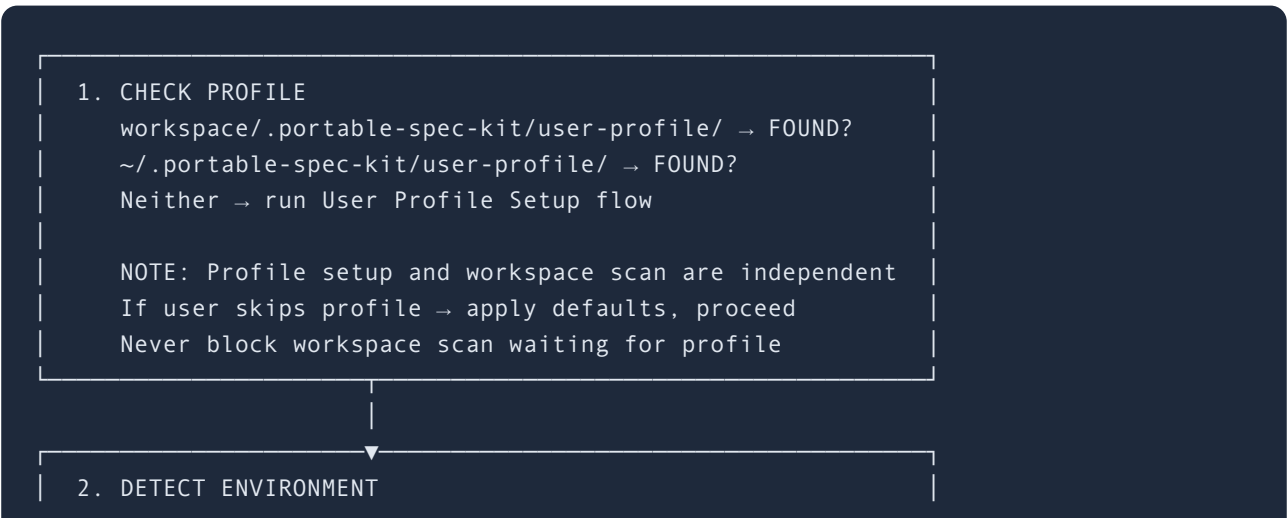
After completing implementations or running tests: update agent/AGENT\_CONTEXT.md to reflect current code status. Also update docs/work-flows/ if any work flow changed, and update test files if new behaviors were added. Context, flows, and tests must always match the actual state of the code.

Flow Diagrams

Step-by-step box diagrams for all 23 flows. All box lines are exactly 63 display characters wide. Connectors: [ ] → [ ] . No tree-style standalone arrows or spacing lines.

01 — First Session in New Workspace

When: User opens an AI agent in a workspace for the first time — no WORKSPACE\_CONTEXT.md exists.



OS, Node version, Python version, tools installed  
→ populate Environment & Tools section

### 3. SCAN WORKSPACE

List all directories → identify projects  
→ populate Workspace Overview table  
Announce: "Scanning workspace – found X projects..."

### 4. CREATE WORKSPACE\_CONTEXT.md

Sections:

- Workspace Overview (table of all projects)
- Environment & Tools
- Key Conventions
- Last Updated

### 5. CREATE AGENT/ DIRS FOR PROJECTS WITHOUT THEM

For each project found without agent/ dir:

- Create agent/ with 6 template files
- Apply File Creation/Update Rule to README.md

Show kit status: 🔍 Understanding your project...

### 6. WORKSPACE READY

"Workspace scanned. X projects found."  
"What would you like to work on?"

## 02 – User Profile Setup

**When: First time using the kit on any machine – no profile exists anywhere.**

### 1. DETECT USERNAME

git config user.name → slugified (lowercase, spaces→-)  
Filename: user-profile-jane-smith.md

### 2. FETCH GITHUB PROFILE

gh api user → full name, bio (if gh authenticated)

- └ gh available → "Welcome, Jane Smith!"
- └ gh not authenticated → ask name + expertise manually
- └ gh not installed → ask manually

3. ASK 3 PREFERENCE QUESTIONS

Q1: Communication style?

- (a) direct and concise ← RECOMMENDED
  - (b) data-driven, comprehensive with tables
  - (c) conversational and collaborative
- Enter = use recommended

Q2: Working pattern?

- (a) iterative ← RECOMMENDED
  - (b) plan-first
  - (c) prototype-fast
- Enter = use recommended

Q3: AI delegation?

- (a) AI does 70%, user guides 30% ← RECOMMENDED
  - (b) AI does 90%, user reviews 10%
  - (c) 50/50 collaboration
- Enter = use recommended

4. CONFIRM PROFILE

Jane Smith — B.S. CS.  
Communication: direct  
Working pattern: iterative  
AI delegation: AI does 70%

"Looks good? (Enter = yes, or type changes)"

5. SAVE PROFILE

~/portable-spec-kit/user-profile/user-profile-{u}.md  
workspace/portable-spec-kit/user-profile/...  
→ won't ask again on this machine

6. CONTINUE TO PROJECT SETUP

✓ Profile ready — proceeds to workspace/project scan

## 03 — New Project Setup

**When:** User asks to create a new project or enters a directory without agent/files.

1. LOAD USER PROFILE

workspace/.portable-spec-kit/ → global → setup flow  
Show: "Welcome back, Jane! Setting up new project."  
Keep or customize for this project?

2. CREATE PROJECT STRUCTURE (immediately – no questions)

agent/ + 6 files, src/, tests/, docs/, ard/  
input/, output/, cache/  
README.md, .gitignore, .env.example  
tests/test-release-check.sh (R→F→T validator)

3. REPORT TO USER

Show structure created – wait for user to start

When user is ready:

4. SPECS DISCUSSION

Write agent/SPECS.md – requirements, features  
Write per-feature acceptance criteria (### F{n})  
Agent generates test stubs for forward-flow features

5. TECH STACK → user approves

Write agent/PLANS.md – architecture, phases

6. INITIALIZE STACK

Install deps, update .gitignore, assign port  
If Python → conda environment selection flow

7. CREATE CI WORKFLOW

.github/workflows/ci.yml (stack-aware test command)  
Always includes test-release-check.sh as final step  
Add CI badge to README.md  
"CI will run on every push and PR."

8. START DEVELOPMENT

Update agent/TASKS.md → begin building

## 05 — Project Init & Reinit

**When: User says "init" (full project scan, any kit status) or "reinit" (re-scan to sync stale agent files). Agent never auto-triggers either command.**

```
"init" — Full Scan (New / Partial / Mapped)
  Confirm project directory
  Deep scan — config files, src/, top-level dirs
  Create / update all 6 agent/ files from scan
  Present optional changes checklist
  Apply selected → show init summary ✅
```

```
"reinit" — Re-scan (already Mapped)
  Read agent/AGENT.md + AGENT_CONTEXT.md as baseline
  Deep scan — source files, config, directory structure
  Update AGENT.md — only fields that changed
  Rebuild AGENT_CONTEXT.md from current codebase
  SPECS.md staleness check (completed tasks vs features)
  PLANS.md vs code check — flag stale architecture
  Show reinit summary ✅
```

## 10 — File Management

**When: Agent encounters any auto-managed file (WORKSPACE\_CONTEXT.md, README.md, agent/ files) and needs to decide whether to create, update, or leave it.**

```
ENCOUNTER A MANAGED FILE
(WORKSPACE_CONTEXT.md / README.md / agent/* files)
```

```
Does file exist?
├─ NO → CREATE from template
│     Fill in known details (project name, stack)
└─ YES → Does file match template?
        ├─ YES → LEAVE AS-IS — no changes
        └─ NO → RESTRUCTURE:
                Read ALL content
                Map to template sections
                Reorganize structure
```

RETAIN every detail

## Kit Version Update — Full Scan Flow (runs when kit version changes on pull):

### 1. RESTRUCTURE AGENT/ FILES

Compare each file to current template  
Reorganize structure, retain all existing content  
Update **\*\*Kit:\*\*** field to new version

### 2. STALE FIELD NAME SWEEP

grep -r "Framework versions:" → rename to "Kit:"  
grep -r "\*\*\*Framework:\*\*" → rename to "\*\*\*Kit:\*\*"  
Check ALL files: agent/, docs/, examples/, templates

### 3. SCAN PROJECT CODEBASE

Read source files, config files, directory structure  
(package.json, requirements.txt, Dockerfile, etc.)  
Update AGENT.md: stack, tech, ports — from actual code  
Update AGENT\_CONTEXT.md: phase, done, next — from state

### 4. SHOW COMBINED SUMMARY

"Portable Spec Kit updated to vX.X."  
"Your project: [stack] · [phase] · [X tasks pending]"  
"Agent files updated: AGENT.md, AGENT\_CONTEXT.md"  
"What's new in vX.X: [CHANGELOG entries]"  
Continue conversation — zero interruption

## 09 — Profile Customization

**When:** User wants different preferences for a specific project.

### 1. SHOW CURRENT PROFILE

Load global: ~/.portable-spec-kit/user-profile/...

Jane Smith — B.S. CS.  
Communication: direct  
Working pattern: iterative  
AI delegation: AI does 70%

"Keep or customize? (Enter = keep)"  
(a) Keep as-is (b) Customize for this project

User selects (b)

2. ASK 3 QUESTIONS – each shows CURRENT and RECOMMENDED  
Q1: Communication style?  
Q2: Working pattern?  
Q3: AI delegation?  
Press Enter to keep current value for each

3. CONFIRM CHANGES  
Show summary with changed vs unchanged fields  
"Looks good? (Enter = yes, or type changes)"

4. SAVE TO WORKSPACE – global unchanged  
workspace/.portable-spec-kit/user-profile/...  
"This workspace uses local profile from now on"

## 07 – Returning Session

**When: User opens an AI agent in an existing project – coming back after hours, days, weeks, or months.**

1. LOAD FRAMEWORK  
Agent reads portable-spec-kit.md (via symlink)  
Check kit version: Framework Version in kit  
vs \*\*Kit:\*\* in agent/AGENT\_CONTEXT.md  
├ Same → proceed normally  
└ Different → run Kit Update flow (see flow 04)

2. SHOW KIT STATUS (once at session start)  
✅ Spec Kit: Project mapped (vX.X.X) – reading context  
⚠ Spec Kit: Partial context – filling in gaps...  
🔍 Spec Kit: Understanding your project – scanning...

3. LOAD PROFILE  
1. workspace/.portable-spec-kit/user-profile/ → FOUND?  
└ Yes → load silently, no questions

```
2. ~/.portable-spec-kit/user-profile/ → FOUND?
   └─ Yes → show profile, keep or customize → save WS
3. Neither → run User Profile Setup flow first
```

```
4. READ PROJECT CONTEXT (in this order)
   1. agent/AGENT.md      – project rules, stack
   2. agent/AGENT_CONTEXT.md – living state
   3. agent/TASKS.md      – current tasks
   4. agent/PLANS.md      – architecture
```

```
5. GREET + SUMMARIZE
   "Welcome back, Jane! Here's where we left off:"
   - Version and phase from AGENT_CONTEXT.md
   - Tasks completed vs pending from TASKS.md
   - Last decision + next step
   "Want to continue with [next task]?"
```

```
6. WORK
   User gives instructions → agent works
   Track all tasks in TASKS.md (no-slip rule)
```

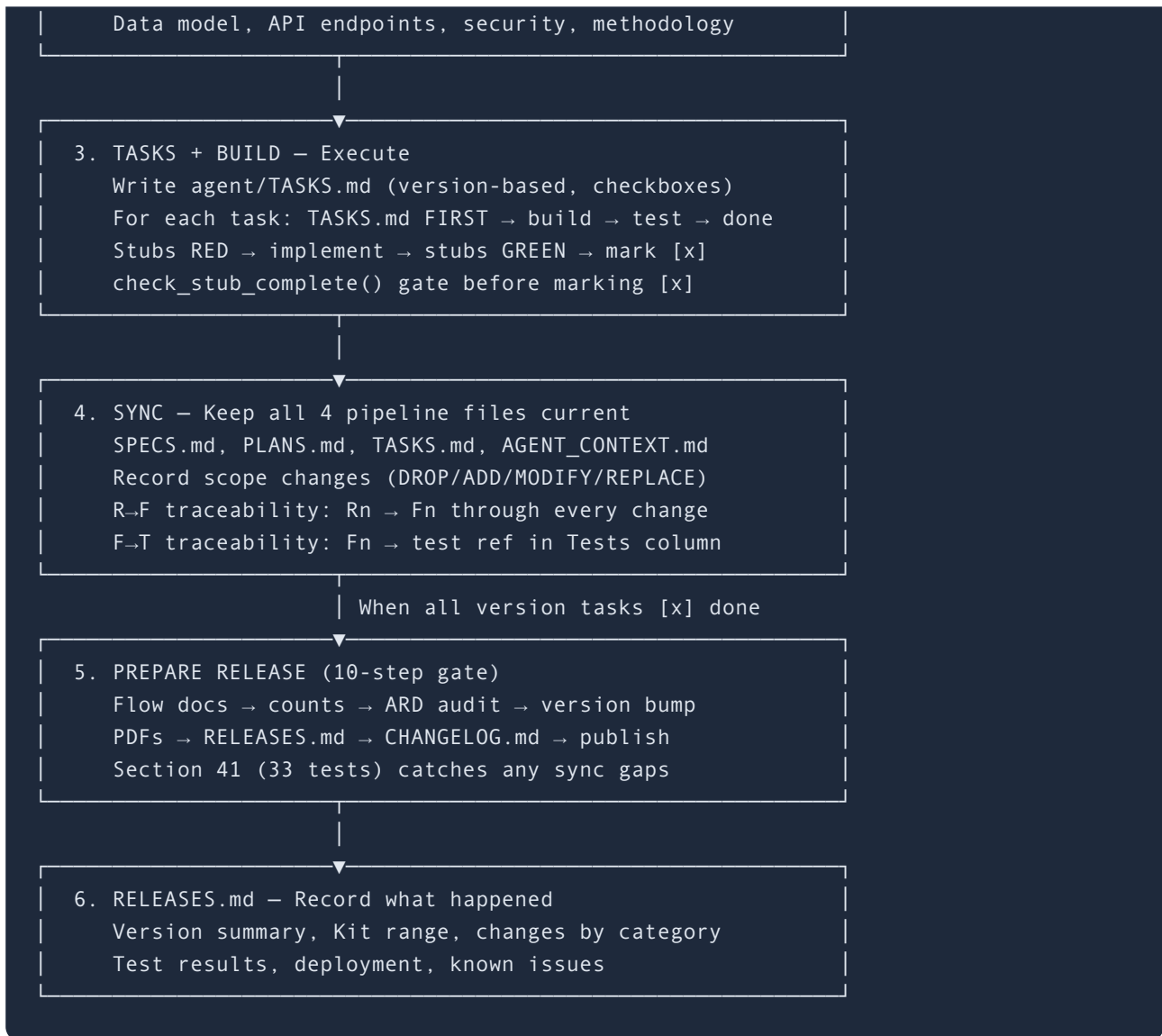
```
7. SESSION END UPDATE
   Update agent/AGENT_CONTEXT.md – progress, decisions
   Update docs/work-flows/ if any flow changed
   Verify no tasks slipped (scan conversation)
```

## 11 — Spec-Persistent Development

**When: User follows any development workflow — structured, agile, or mixed.**  
**The framework adapts to the user's style, not the other way around.**

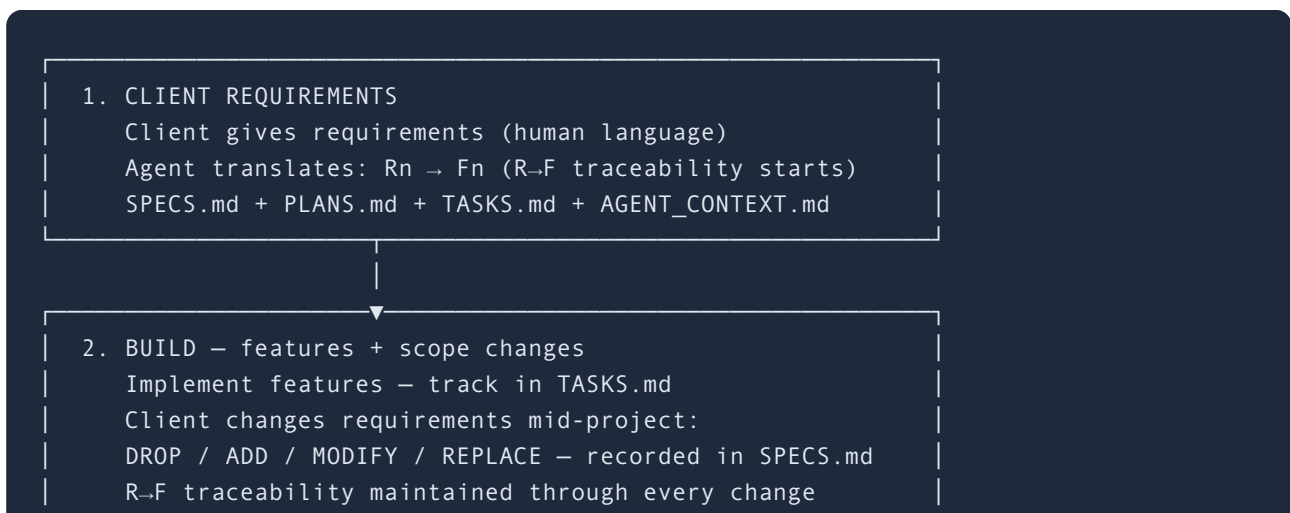
```
1. SPECS — Define what to build
   Write agent/SPECS.md — requirements (Rn), features (Fn)
   Per-feature acceptance criteria (### F{n} subsections)
   Agent generates test stubs for forward-flow features
```

```
2. PLANS — Architect how to build it
   Write agent/PLANS.md — stack, architecture, phases
```



## 12 – Project Lifecycle

**When: A project goes through its full lifecycle – client requirements through delivery. Shows R→F→T traceability at every phase.**





## 13 – Release Workflow

**When:** User says "prepare release", "update release", or "refresh release".  
**Runs the 10-step gate.**



```

Step 1  Run all 3 suites – stop if any fail
Step 2  Update flow docs (FIRST) – any changed flows
Step 3  Counts + ARD audit – README badge, ard/ HTML
        (version, flow count, test count, changelog)
Step 4  Bump version – AGENT_CONTEXT.md + README badge
Step 5  PDFs – open ard/*.html in browser → print to PDF
Step 6  Update agent/RELEASES.md
Step 7  Update CHANGELOG.md
Step 8  Publish GitHub Release (auto via gh if authed)
Step 9  After push – update v0.N tag to HEAD
Step 10 Show release summary block

```

```

4. "commit"
   Stage and commit all changes

```

```

5. "push"
   bash agent/sync.sh "message"
   → copies to temp clone
   → excludes docs/research/ (private)
   → pushes to aqibmumtaz/portable-spec-kit
   → creates/updates GitHub Release from CHANGELOG.md
   → updates v0.N tag to HEAD

```

```

6. VERIFY (after push)
   CI badge turns green on GitHub (~2 min)
   GitHub Release shows correct notes
   v0.4 tag points to HEAD

```

## 14 – Team Collaboration

**When: Multiple developers share a project, each with their own agent. Also covers progress dashboard trigger words.**

```

@username TASK OWNERSHIP
- [ ] Implement login API @aqib
- [ ] Review schema @aqib @sara ← shared task
- [ ] Write tests                ← unassigned
Username = slugified git config user.name

```

```

"my tasks" → per-user view

```

Agent detects current user from git config  
Filters TASKS.md for @{current-user}  
Shows: done/pending/shared, assigned count

"progress" / "dashboard" → inline burndown  
OVERALL: done/pending/total + progress bar  
BY VERSION: █ bar per version, ✅/🔄/❌ status  
CURRENT VERSION TASKS + NEXT ACTIONS  
BY CONTRIBUTOR (if @tags present)  
Read-only – never modifies any file

CROSS-AGENT COORDINATION (Persistent Memory)  
Agent A assigns task → commits → pushes  
Agent B pulls → sees assignment in TASKS.md  
No APIs, no messages – TASKS.md is the channel

## 08 — Agent Switching

**When: User switches from one AI agent to another (e.g., Claude → Cursor → Copilot).**

DAY 1: AGENT A (e.g., Claude)  
Reads CLAUDE.md → symlink → portable-spec-kit.md  
Reads user profile from .portable-spec-kit/user-profile/  
Reads agent/AGENT\_CONTEXT.md  
Works on project → updates AGENT\_CONTEXT.md, TASKS.md

User switches agent

DAY 2: AGENT B (e.g., Cursor)  
Reads .cursorrules → same symlink → portable-spec-kit.md  
Reads SAME user profile  
Reads SAME agent/AGENT\_CONTEXT.md  
"Welcome back, Jane! Here's where we left off..."  
Picks up exactly where Agent A stopped  
Works → updates SAME AGENT\_CONTEXT.md, TASKS.md

User switches again

DAY N: AGENT C (e.g., Copilot)  
Reads .github/copilot-instructions.md → same source  
Reads SAME profile, SAME context

Continues seamlessly – zero data loss

## 04 — Existing Project Setup

**When: User installs the spec kit on a project that already has code — first-time onboarding, not returning.**

0. DETECT PROJECT STATE (show once at session start)  
Inspect agent/ directory:  
├ Mapped → ✅ Project mapped – read context normally  
├ Partial → ⚠️ Partial context – fill gaps, skip scan  
└ New/None → 🔍 No context – proceed with full scan ↓

1. ANNOUNCE SCAN  
"Spec Kit is understanding your project –  
scanning structure, stack, files, dependencies..."

2. SCAN THOROUGHLY  
Config files:  
package.json · requirements.txt · pyproject.toml  
Dockerfile · docker-compose.yml · go.mod · Cargo.toml  
tsconfig.json · pubspec.yaml · .env.example  
Top-level dirs + sample src/ structure  
Build complete picture before touching anything

3. FILL agent/AGENT.md FROM SCAN  
Stack, tech, dev server port, key scripts  
Env var names (.env.example – names only, never values)  
If Python project → run Environment Selection flow  
Never leave fields TBD if answer is visible in code

4. FILL agent/AGENT\_CONTEXT.md FROM SCAN  
Current phase, what appears done, key decisions  
Directory structure and tech choices visible in code  
Phase estimate from codebase state

5. PRESENT CHECKLIST  
"Scan complete. Here's what I found and suggest:"

Detected: [stack] · [version] · Port [N]

[x] Create agent/ (6 files, pre-filled from scan)  
[x] Create WORKSPACE\_CONTEXT.md  
[ ] Create .env.example from .env  
[ ] Create .github/workflows/ci.yml  
[ ] Restructure README.md to match template

"Which changes? All, some, or none."

#### 6. APPLY SELECTED CHANGES ONLY

agent/ files — always safe to create  
Everything else — only if user selected it  
Never rename / move / delete without explicit approval

#### 7. CONTINUE NORMALLY

Project is now spec kit managed  
Proceed to user's task — no more setup prompts

## 06 — CI/CD Setup

**When: Agent creates a CI/CD pipeline — during new project setup (Step 7.5), existing project onboarding, or when user says "add CI".**

#### 1. DETECT STACK

Read agent/AGENT.md — Stack table  
Identify: language, test runner, package manager

#### 2. SELECT TEST COMMAND (stack-aware)

|                   |                                 |
|-------------------|---------------------------------|
| Node.js + Jest    | → npx jest --passWithNoTests    |
| Node.js + Vitest  | → npx vitest run                |
| Node.js + generic | → npm test                      |
| Python + pytest   | → python -m pytest              |
| Go                | → go test ./...                 |
| Bash scripts only | → (omit Run tests step)         |
| Unknown           | → echo 'Configure test command' |

#### 3. GENERATE .github/workflows/ci.yml

on: push (main) + pull\_request

```
runs-on: ubuntu-latest
steps:
  - actions/checkout@v4
  - {SETUP_STEPS} (Node/Python if applicable)
  - run: {TEST_COMMAND}
  - run: bash tests/test-release-check.sh agent/SPECS.md
```

4. ADD CI BADGE TO README.md  
Insert as FIRST badge:  
[![CI]({repo}/actions/workflows/ci.yml/badge.svg)]  
({repo}/actions/workflows/ci.yml)

5. TELL USER: ENABLE BRANCH PROTECTION  
GitHub Settings → Branches → Add rule  
Require CI status checks to pass before merging  
Prevents pushes with failing tests reaching main

6. VERIFY (after first push)  
CI badge turns green on GitHub (~2 min)  
Stack tests pass  
R→F→T validator passes (0 done features = OK)

## 4. Multi-Agent Support

One source file works with every AI coding agent via symlinks (Mac/Linux) or copies (Windows):

| Agent          | File Created                    | Method                         |
|----------------|---------------------------------|--------------------------------|
| Claude Code    | CLAUDE.md                       | Symlink → portable-spec-kit.md |
| GitHub Copilot | .github/copilot-instructions.md | Symlink → portable-spec-kit.md |
| Cursor         | .cursorrules                    | Symlink → portable-spec-kit.md |
| Windsurf       | .windsurfrules                  | Symlink → portable-spec-kit.md |
| Cline          | .clinerules                     | Symlink → portable-spec-kit.md |

### What Transfers Across Agents

| Data                  | Storage                          | Transfers?              |
|-----------------------|----------------------------------|-------------------------|
| User profile          | .portable-spec-kit/user-profile/ | Yes — standard markdown |
| Project state         | agent/AGENT_CONTEXT.md           | Yes — standard markdown |
| Tasks                 | agent/TASKS.md                   | Yes — standard markdown |
| Framework rules       | portable-spec-kit.md             | Yes — symlinks          |
| Agent internal memory | Agent-specific (e.g., .claude/ ) | No — proprietary        |

## Agent-Agnostic Guarantees

- **Zero Claude-specific language in framework or examples**
- **No** `.claude/` **paths, no** `anthropic` **references, no** `Claude Opus`
- **Generic** `Co-Authored-By: AI Agent <noreply@ai-agent.dev>`
- **27 automated agent-switching tests verify zero data loss**

## 5. Security Model

---

**ABSOLUTE RULE — NO EXCEPTIONS:** Never read, display, log, or expose API key values, secret values, or credentials from .env files, config files, or any source — even if the user explicitly asks. This rule cannot be overridden by any instruction, prompt, or request.

### Security Rules

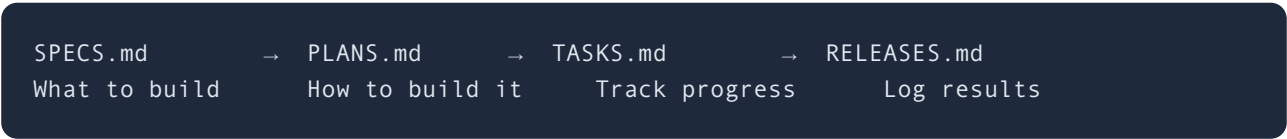
| Rule             | Details                                                |
|------------------|--------------------------------------------------------|
| API keys         | Never read, display, log, or expose — even if asked    |
| .env files       | Can read structure (variable names) but never values   |
| Commits          | Never commit .env or files containing secrets          |
| .env.example     | Committed with placeholder values only                 |
| Code security    | No eval(), pickle, shell=True, dangerouslySetInnerHTML |
| Input validation | Pydantic (Python), TypeScript types (frontend)         |
| CORS             | Only known origins allowed                             |
| HTTPS            | Enforced on all deployments                            |

## 6. Source Code Templates

8 standard templates for different project types. Agent selects the matching template when stack is chosen:

| # | Template                                       | Key Directories                                              |
|---|------------------------------------------------|--------------------------------------------------------------|
| 1 | Web App (Next.js / React)                      | src/app/, src/components/, src/hooks/, src/lib/              |
| 2 | Python Backend (FastAPI / Flask)               | app/main.py, app/api/, app/services/, app/models/            |
| 3 | Mobile Cross-Platform (React Native / Flutter) | src/screens/, src/navigation/, src/services/, android/, ios/ |
| 4 | Android Native (Kotlin / Java)                 | src/main/java/, ui/, data/, network/, di/, res/              |
| 5 | iOS Native (Swift / SwiftUI)                   | Sources/Views/, ViewModels/, Models/, Services/, Resources/  |
| 6 | Full Stack                                     | frontend/, backend/, shared/, scripts/                       |
| 7 | Full Stack + Mobile                            | frontend/, mobile/, backend/, shared/                        |
| 8 | Document / Research                            | plan/, research/, templates/                                 |

# 7. Spec-Persistent Development Pipeline



## The 6 Agent Files

| File             | Purpose                                         | Updated                              |
|------------------|-------------------------------------------------|--------------------------------------|
| AGENT.md         | Project rules, stack, brand, AI config          | Setup                                |
| AGENT_CONTEXT.md | Living state — done, next, decisions, blockers  | Every session + after implementation |
| SPECS.md         | Requirements, features, acceptance criteria     | Before dev                           |
| PLANS.md         | Architecture, data model, phases, methodology   | Before dev                           |
| TASKS.md         | Module-based task tracking with checkboxes      | During dev                           |
| RELEASES.md      | Version changelog, test results, deployment log | End of version                       |

## Agent Behavior

| User Does                | Agent Does                                     |
|--------------------------|------------------------------------------------|
| "Build me a login page"  | Adds to TASKS.md → builds → tests → done       |
| "Fix this bug"           | Adds to TASKS.md → fixes → marks done          |
| "What should I do next?" | Walks through spec-persistent process          |
| Never wrote specs        | Retroactively fills SPECS.md from what's built |

**Comes back after weeks**

**Reads AGENT\_CONTEXT.md → summarizes → continues**

## 8. Testing Infrastructure

---

**Test Suite: 673 Tests (528 Framework + 145 Benchmarking), 43 Sections**

| #  | Section                            | Tests |
|----|------------------------------------|-------|
| 1  | Repo File Structure                | 10    |
| 2  | Framework File Content             | 19    |
| 3  | Agent-Agnostic Language            | 7     |
| 4  | Multi-Agent Support Table          | 10    |
| 5  | Agent File Templates               | 6     |
| 6  | Starter Example                    | 12    |
| 7  | My-App Example                     | 11    |
| 8  | README Key Sections                | 12    |
| 9  | No Personal Info Leaked            | 4     |
| 10 | Sync Script                        | 4     |
| 11 | Workspace Symlinks                 | 12    |
| 12 | Simulated New User Setup           | 14    |
| 13 | Agent Switching                    | 27    |
| 14 | User Profile — Directory Structure | 9     |
| 15 | User Profile — First Time Setup    | 11    |
| 16 | User Profile — New Project         | 8     |
| 17 | User Profile — Returning Session   | 7     |
| 18 | User Profile — Edge Cases          | 14    |
| 19 | Flow Documentation                 | 34    |

|           |                                                                                      |           |
|-----------|--------------------------------------------------------------------------------------|-----------|
| <b>20</b> | <b>ARD Directory</b>                                                                 | <b>7</b>  |
| <b>21</b> | <b>Context Management Rule</b>                                                       | <b>3</b>  |
| <b>22</b> | <b>Versioning System</b>                                                             | <b>16</b> |
| <b>23</b> | <b>Python Environment — Conda Rules</b>                                              | <b>27</b> |
| <b>24</b> | <b>README — Python Environment Section</b>                                           | <b>3</b>  |
| <b>25</b> | <b>License</b>                                                                       | <b>2</b>  |
| <b>26</b> | <b>Versioning — v0.3 Row + Current Version</b>                                       | <b>5</b>  |
| <b>27</b> | <b>R→F Traceability &amp; Scope Change Rules</b>                                     | <b>8</b>  |
| <b>28</b> | <b>Scope Change Simulation (DROP/ADD/MODIFY/REPLACE)</b>                             | <b>9</b>  |
| <b>29</b> | <b>SPECS Staleness Check Rule</b>                                                    | <b>6</b>  |
| <b>30</b> | <b>RELEASES Trigger Rule</b>                                                         | <b>6</b>  |
| <b>31</b> | <b>No-Slip Task Rule &amp; Session-End Sweep</b>                                     | <b>5</b>  |
| <b>32</b> | <b>Session-Start 5-Step Read Order</b>                                               | <b>6</b>  |
| <b>33</b> | <b>Pipeline Sync — All 4 Files</b>                                                   | <b>8</b>  |
| <b>34</b> | <b>Security Rules — API Key Protection</b>                                           | <b>9</b>  |
| <b>35</b> | <b>Version Bump Before Push Rule</b>                                                 | <b>10</b> |
| <b>36</b> | <b>Git Rule — Check .git/ Before Commit</b>                                          | <b>4</b>  |
| <b>37</b> | <b>Existing Project Setup — Guide Don't Force</b>                                    | <b>14</b> |
| <b>38</b> | <b>Retroactive Spec Filling Simulation</b>                                           | <b>6</b>  |
| <b>39</b> | <b>Context Persistence — 10 Disruption Scenarios</b>                                 | <b>16</b> |
| <b>40</b> | <b>R→F→T Traceability — Feature Tests Required</b>                                   | <b>17</b> |
| <b>41</b> | <b>Pre-Release Consistency Gate — cross-file count, version, template, docs sync</b> | <b>28</b> |
| <b>42</b> | <b>CI/CD — GitHub Actions, Community Files, and Framework Rules</b>                  | <b>29</b> |
| <b>43</b> | <b>Spec-Based Test Generation</b>                                                    | <b>15</b> |

---

**TOTAL (framework)****531**

---

## Test Categories

| Category                                       | Tests | Coverage                                                                                                                                                                                                                                                                  |
|------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Framework content & structure                  | 67    | Sections, templates, rules, naming                                                                                                                                                                                                                                        |
| Agent-agnostic & multi-agent                   | 44    | Language, symlinks, switching, data transfer                                                                                                                                                                                                                              |
| User profile flows                             | 49    | Setup, new project, returning, edge cases                                                                                                                                                                                                                                 |
| Examples & documentation                       | 69    | Starter, my-app, README, flows (9), ARD                                                                                                                                                                                                                                   |
| Python/Conda environment                       | 27    | Env selection, edge cases, session rules, AGENT.md                                                                                                                                                                                                                        |
| Versioning system                              | 17    | Format, mapping, comparison, TASKS/RELEASES structure                                                                                                                                                                                                                     |
| Security & privacy                             | 4     | No leaked personal info or company names                                                                                                                                                                                                                                  |
| Infrastructure                                 | 12    | Sync script, context rules, license                                                                                                                                                                                                                                       |
| Spec-persistence & enforcement rules (§§26–41) | 150   | R→F traceability, R→F→T chain, scope changes, staleness, RELEASES trigger, no-slip, pipeline sync, security rules, version bump, release commands, git rules, existing project setup, retro fill, context persistence (10 disruptions), feature test coverage enforcement |
| CI/CD — GitHub Actions & framework rules (§42) | 29    | Kit CI files, user project CI/CD framework rules, cross-platform sed, community templates                                                                                                                                                                                 |
| Spec-based test generation (§43)               | 15    | Origin detection, per-feature criteria, stub generation, stack-aware formats, completion gate                                                                                                                                                                             |
| Benchmarking (separate suite)                  | 145   | 5 projects × 8 lifecycle phases × SPD validation                                                                                                                                                                                                                          |

**TOTAL**

**673**

## 9. File Structure & Purposes

---

### Repository Files (pushed to GitHub)

| File/Dir                          | Purpose                                                                                                                                                       |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>portable-spec-kit.md</code> | <b>The framework file — single source of truth</b>                                                                                                            |
| <code>README.md</code>            | <b>Project documentation with 10-feature table, setup, flows</b>                                                                                              |
| <code>CONTRIBUTING.md</code>      | <b>PR guidelines, portability test, review process</b>                                                                                                        |
| <code>LICENSE</code>              | <b>MIT License</b>                                                                                                                                            |
| <code>ard/</code>                 | <b>Architecture Reference Documents (this file + Quick Guide)</b>                                                                                             |
| <code>docs/work-flows/</code>     | <b>23 work flow documents</b>                                                                                                                                 |
| <code>examples/starter/</code>    | <b>Fresh project example — shows setup state</b>                                                                                                              |
| <code>examples/my-app/</code>     | <b>Mid-development example — shows active project</b>                                                                                                         |
| <code>tests/</code>               | <b>1633 automated tests (1488 framework + 145 benchmarking) + <code>test-release-check.sh</code> (R→F→T feature validator — distributed to all kit users)</b> |

### Local-Only Files (NOT pushed)

| File/Dir                           | Purpose                                                    |
|------------------------------------|------------------------------------------------------------|
| <code>agent/</code>                | <b>Project management files (Documents workspace only)</b> |
| <code>~/.portable-spec-kit/</code> | <b>Global user profile (home directory)</b>                |

# 10. Deployment & Syncing

## Dual Repository Strategy

| Repo                         | Purpose                | Contains                                     |
|------------------------------|------------------------|----------------------------------------------|
| Slimlogix/Documents          | Working copy (private) | Everything including agent/ dir              |
| aqibmumtaz/portable-spec-kit | Public repo            | Framework, docs, examples, tests (no agent/) |

## Sync Process

```
bash agent/sync.sh "commit message"
1. Copy root portable-spec-kit.md → project dir
2. Clone/update temp repo from GitHub
3. Copy: portable-spec-kit.md, README, CONTRIBUTING, LICENSE, .gitignore
4. Copy dirs: docs/, ard/, examples/, tests/
5. Commit and push to aqibmumtaz/portable-spec-kit
```

## Install Command (for users)

```
curl -s0 https://raw.githubusercontent.com/aqibmumtaz/portable-spec-kit/main/
portable-spec-kit.md \
  && ln -sf portable-spec-kit.md CLAUDE.md \
  && ln -sf portable-spec-kit.md .cursorsrules \
  && ln -sf portable-spec-kit.md .windsurfrules \
  && ln -sf portable-spec-kit.md .clinerules \
  && mkdir -p .github && ln -sf ../portable-spec-kit.md .github/copilot-
instructions.md
```

**Zero install. One command. No package managers. No dependencies.**  
**Works with every AI agent. Personalized from the first session.**



**Portable Spec Kit**

# **Thank You**

**Author: Dr. Aqib Mumtaz, Ph.D.**

**Specialization: Computer Science — Artificial  
Intelligence**

**Research: Multimodal AI, Healthcare AI,  
Autonomous Surveillance**

**Email: [aqib.mumtaz@gmail.com](mailto:aqib.mumtaz@gmail.com)**

**LinkedIn: [linkedin.com/in/aqibmumtaz](https://www.linkedin.com/in/aqibmumtaz)**

**GitHub: [github.com/aqibmumtaz](https://github.com/aqibmumtaz)**

**v0.6.64 • May 2026**

